Reflections and Coxeter Relations in Euclidean Geometry

A Lean Formalization Supplement

Jurek Różański

Military University of Technology, Warsaw, Poland

CGJteamLab

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Contents

Chapter 1

Line Reflection as a Hilbert Generator

1.1 Overview

This chapter develops line reflection as a genuine transformation derived from the synthetic Hilbert foundation. The corresponding Lean module is

```
CGJteamLab/Coxeter/Reflection.lean
```

The module is deliberately separated from the flat sequence of Euclid Proposition files. It is not another proposition of the *Elements*; it is a transformation layer derived over the same Hilbert foundation.

The immediate objective is to obtain a genuine reflection generator

$$r_l : \mathrm{Point} \longrightarrow \mathrm{Point}$$

without introducing reflection as a new primitive geometric operation. The construction starts with a geometric relation, proves existence and uniqueness, and only then packages the relation as a point transformation.

1.2 Coxeter Motivation

Coxeter and Moser begin with generators and defining relations and later specialize to groups generated by reflections. In the reflection-group language, each basic reflection is involutory:

$$r_i^2 = 1,$$

while the geometry of pairs of reflecting hyperplanes is encoded by the periods of products

$$(r_i r_j)^{p_{ij}} = 1.$$

Their graphical notation records these pairwise periods by a Coxeter diagram.

The present module addresses only the first relation. It constructs a single Hilbert line reflection and proves, in literal operator form,

$$r_l(r_l(P)) = P.$$

Products of distinct reflections and relations of the form $(r_l r_m)^p = 1$ belong to the next layer and are intentionally absent from this file.

The relevant source locations in Coxeter–Moser, *Generators and Relations for Discrete Groups*, second edition, are Sections 1.1, 4.3, and 9.1–9.3.

1.3 Geometric Design

1.3.1 Why the construction begins as a relation

The Hilbert development does not take a global transformation space as a primitive object. The natural first notion is therefore a relation between a point P and its reflected point P'.

For a point off the axis l, reflection is characterized by a foot H satisfying

$$H \in l, \qquad P - H - P', \qquad PH \cong HP',$$

together with the condition that PH is perpendicular to the axis.

For a point on the axis, the reflected point is the point itself.

Thus the geometric core is

$$\boxed{P' = P \text{ on } l \quad \text{or} \quad H \in l,\ H = \mathrm{mid}(P, P'),\ PH \perp l.}$$

1.3.2 ReflectionAxis

The Hilbert API often represents a line geometrically by two distinct points known to lie on it. The reflection axis therefore stores both the carrier and witnesses sufficient for perpendicular constructions.

```
structure ReflectionAxis
    [HilbertIncidence Geo] where
  carrier : Geo.Line
  A : Geo.Point
  B : Geo.Point
  hAB : Not (A = B)
  hA : HilbertIncidence.OnLine A carrier
  hB : HilbertIncidence.OnLine B carrier
```

This is not additional geometry. It packages the nondegenerate line data already required by the existing Hilbert construction interface.

1.3.3 PerpendicularToAxis

Perpendicularity to the axis is represented using the established point-based right-angle language.

```
def PerpendicularToAxis
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
```

```
    (H P : Geo.Point) : Prop :=
  And
    (HilbertIncidence.OnLine H axis.carrier)
    (And
      (Not (HilbertIncidence.OnLine P axis.carrier))
      (Exists fun R : Geo.Point =>
        And
          (HilbertIncidence.OnLine R axis.carrier)
          (And
            (Not (R = H))
            (HilbertRightAngle Geo R H P))))
```

The witness R supplies a nondegenerate direction of the carrier. No primitive relation "two lines are perpendicular" is added.

1.3.4 IsLineReflection

The reflection relation has a fixed-point branch and an off-axis branch.

```
def IsLineReflection
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P P' : Geo.Point) : Prop :=
  Or
    (And
      (HilbertIncidence.OnLine P axis.carrier)
      (P' = P))
    (And
      (Not (HilbertIncidence.OnLine P axis.carrier))
      (Exists fun H : Geo.Point =>
        And
          (PerpendicularToAxis Geo axis H P)
          (HilbertIsMidpoint Geo H P P')))
```

The midpoint condition packages both strict betweenness $P - H - P'$ and equality of the two half-segments.

1.4 Existence of the Reflected Point

The first main theorem is

```
theorem line_reflection_exists
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P : Geo.Point) :
    exists P' : Geo.Point,
      IsLineReflection Geo axis P P'
```

The proof splits on whether P lies on the axis.

If $P \in l$, choose $P' = P$.

If $P \notin l$, Euclid I.12 in its Hilbert reconstruction supplies the perpendicular foot H. The established theorem

```
hilbert_extend_segment_beyond
```

then extends PH beyond H to a point P' satisfying

$$P - H - P', \qquad PH \cong HP'.$$

This is exactly the midpoint condition needed by `HilbertIsMidpoint`.

The conceptual dependency is therefore

$$\boxed{\text{perpendicular construction} + \text{equal segment extension} \Longrightarrow \text{existence of reflection.}}$$

1.5 Relational Involution

Before constructing a function, the development proves that the reflection relation is symmetric:

```
theorem line_reflection_symmetric
    ...
    (hRef : IsLineReflection Geo axis P P') :
    IsLineReflection Geo axis P' P
```

For off-axis points, the same midpoint H is used in the reverse direction. The substantive geometric obligation is to transport the right-angle condition from the ray HP to the opposite ray HP'. This uses the already developed theory of right-angle transport and opposite extensions.

This theorem is the relational precursor of the Coxeter involution relation:

$$P \; R_l \; P' \Longrightarrow P' \; R_l \; P.$$

1.6 Fixed Points

The module records the exact fixed-point set of the reflection:

```
theorem line_reflection_fixed_iff_on_axis
    ...
    (P : Geo.Point) :
    IsLineReflection Geo axis P P <->
      HilbertIncidence.OnLine P axis.carrier
```

Thus

$$r_l(P) = P \quad \Longleftrightarrow \quad P \in l$$

once the functional reflection has been constructed.

The companion theorem `line_reflection_moves_off_axis` states that an off-axis point cannot be fixed.

1.7 Right-Angle Carrier Transport

The uniqueness proof exposed a reusable gap in the permanent Hilbert API: a right angle should not depend on which non-vertex point is chosen on the same carrier of one arm.

Three local theorems were therefore established:

```
coxeter_right_angle_sameRay_first
coxeter_right_angle_opposite_first
coxeter_right_angle_collinear_first
```

The first transports a right angle along the same ray. The second reverses the first arm across the vertex. The third combines the two cases using betweenness trichotomy and gives the carrier-level statement required for perpendicular uniqueness.

This is a useful example of the formal difference between an intuitive statement about perpendicular lines and the point/ray representation used in the Hilbert layer.

1.8 Uniqueness of the Perpendicular Foot

The key geometric lemma is

```
theorem perpendicular_foot_unique
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P H K : Geo.Point)
    (hPerpH : PerpendicularToAxis Geo axis H P)
    (hPerpK : PerpendicularToAxis Geo axis K P) :
    H = K
```

The proof is neutral.

Assume $H \neq K$. Because both H and K lie on the reflection axis, carrier transport converts the two perpendicularity witnesses into right angles

$$\angle KHP \quad \text{and} \quad \angle HKP.$$

Hence the nondegenerate triangle HKP would have right angles at both H and K.

Extend KH beyond H. The resulting exterior angle is again right. All right angles are congruent, so this exterior angle is congruent to the remote interior right angle at K. Hilbert's exterior-angle theorem forbids this. Therefore $H = K$.

The argument uses no Euclidean parallel axiom. Its mathematical core is

$$\boxed{\text{two distinct perpendicular feet} \implies \text{triangle with two right angles} \implies \bot.}$$

1.9 Uniqueness of Reflection

Once the perpendicular foot is unique, uniqueness of the reflected point follows from the existing segment-construction uniqueness theorem.

```
theorem line_reflection_unique
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P P1 P2 : Geo.Point)
    (hRef1 : IsLineReflection Geo axis P P1)
    (hRef2 : IsLineReflection Geo axis P P2) :
    P1 = P2
```

In the off-axis case, both reflections have the same perpendicular foot H. Both points P_1, P_2 lie on the prescribed ray beyond H, and their distances are determined by the same midpoint data. Segment additivity and `hilbert_segment_construction_unique` identify the two endpoints.

Thus

$$\forall P\ \exists!\, P'\ \ \mathrm{IsLineReflection}(l, P, P').$$

This is the logical threshold at which reflection can be promoted from a relation to a point transformation.

1.10 The Reflection Operator

The actual point transformation is selected noncomputably from the existence theorem:

```
noncomputable def lineReflect
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P : Geo.Point) :
    Geo.Point :=
  Classical.choose
    (line_reflection_exists Geo axis P)
```

Its specification theorem is

```
theorem lineReflect_spec
    ...
    (P : Geo.Point) :
    IsLineReflection Geo axis P
      (lineReflect Geo axis P)
```

The use of `Classical.choose` is a packaging step only. The geometric existence and uniqueness were proved before the function was introduced.

1.11 The First Coxeter Relation

The main result of the module is

```
theorem lineReflect_involutive
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
    (P : Geo.Point) :
    lineReflect Geo axis
      (lineReflect Geo axis P) = P
```

The proof has a short conceptual form:

$$\begin{array}{c} P\ R_l\ r_l(P) \\ \Downarrow \quad \text{symmetry} \\ r_l(P)\ R_l\ P \\ r_l(P)\ R_l\ r_l(r_l(P)) \\ \Downarrow \quad \text{uniqueness} \\ P = r_l(r_l(P)). \end{array}$$

Consequently the first Coxeter relation has now been derived inside the Hilbert model:

$$\boxed{r_l^2 = 1.}$$

This is stronger architecturally than merely assuming an abstract involutory generator: the generator itself has been reconstructed from incidence, order, congruence, midpoint, and right-angle geometry.

1.12 Permutation Packaging

The reflection is finally packaged as a permutation of the point set:

```
noncomputable def lineReflectEquiv
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo) :
    Equiv Geo.Point Geo.Point
```

Both the forward and inverse maps are `lineReflect`; the two inverse laws are exactly `lineReflect_involutive`.

The associated simplification lemmas are

```
lineReflectEquiv_apply
lineReflectEquiv_apply_twice
```

This packaging is the interface needed by the next module, where different reflection generators can be composed into words.

1.13 Dependency Summary

The formal architecture may be summarized as follows:

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence}\\
\Downarrow\\
\texttt{HilbertIsMidpoint} \quad + \quad \text{right-angle transport} \quad + \quad \text{segment construction/uniqueness}\\
\Downarrow\\
\texttt{hilbert\_perpendicular\_from\_point\_exists} \quad + \quad \texttt{hilbert\_extend\_segment\_beyond}\\
\Downarrow\\
\texttt{IsLineReflection}\\
\Downarrow\\
\texttt{line\_reflection\_exists} \quad + \quad \texttt{line\_reflection\_symmetric}\\
\Downarrow\\
\texttt{perpendicular\_foot\_unique}\\
\Downarrow\\
\texttt{line\_reflection\_unique}\\
\Downarrow\\
\texttt{lineReflect}\\
\Downarrow\\
\boxed{\texttt{lineReflect\_involutive}}\\
\Downarrow\\
\texttt{lineReflectEquiv}.
\end{gathered}
$$

The only direct Euclid Proposition import required by this module is the Hilbert reconstruction of I.12 for the perpendicular construction. The remaining work belongs to the established Hilbert congruence/order/right angle infrastructure and the local Coxeter helper layer.

1.14 Axiom Audit

The module introduces no new declared geometric axiom and no `sorry`. The uniqueness of the perpendicular foot is proved by neutral angle theory rather than by uniqueness of parallels.

The transformation `lineReflect` is declared `noncomputable` because it uses `Classical.choose` after existence has been established. This should not be confused with a new geometric axiom.

The exact transitive axiom set is obtained with `#print axioms`. At the module level, no new axiom is introduced by `Reflection.lean`.

1.15 Architectural Significance

The construction has the following architecture:

$$
\text{Hilbert geometry} \longrightarrow \text{geometrically defined reflection} \longrightarrow \text{point permutation} \longrightarrow \text{group word.}
$$

This is distinct from two earlier proof mechanisms:

- a Pappus-style relabelling argument, where the same points are read in a different order;

- direct Hilbert congruence proofs, where no transformation object is constructed.

Here the transformation is a first-class object derived from the Hilbert geometry. This is the feature required before Coxeter words and pairwise-generator relations can be studied formally.

1.16 Conclusion

The first Coxeter generator has been reconstructed rather than postulated.

Starting from Hilbert incidence, order, congruence, midpoint, and right-angle geometry, the development proves existence and uniqueness of line reflection, promotes the resulting relation to a permutation of the point set, and establishes

$$\boxed{r_l^2 = 1}.$$

This provides a formal bridge from synthetic Hilbert geometry to a generators-and-relations language of geometric transformations.

Chapter 2

Words in Reflection Generators

2.1 Purpose of the Module

The module

```
CGJteamLab/Coxeter/ReflectionWords.lean
```

is the first layer in which the geometric reflections constructed in `Reflection.lean` are treated as generators of words.

The previous chapter produced, for every reflection axis l, an equivalence

$$r_l : \mathrm{Point} \simeq \mathrm{Point}$$

satisfying

$$r_l^2 = 1.$$

The present module adds no new geometry. Its role is algebraic packaging: it composes reflection generators, iterates their products, states when a pairwise relation is satisfied, and now also records when the displayed exponent is the exact positive period of the product.

Thus the formal transition is

$$\text{geometric reflection} \longrightarrow \text{permutation} \longrightarrow \text{word in two generators.}$$

2.2 Product of Two Reflection Generators

For two axes `axis1` and `axis2`, the basic product is

```
noncomputable def reflectionProduct
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo) :
    Equiv Geo.Point Geo.Point :=
  (lineReflectEquiv Geo axis1).trans
    (lineReflectEquiv Geo axis2)
```

The order convention is important. On points,

$$\text{reflectionProduct}(r_1, r_2)(P) = r_2(r_1(P)).$$

This is recorded by the simplification theorem

```
@[simp]
theorem reflectionProduct_apply
    ...
    (P : Geo.Point) :
    reflectionProduct Geo axis1 axis2 P =
      lineReflect Geo axis2
        (lineReflect Geo axis1 P)
```

Hence `axis1` acts first and `axis2` second.

This convention is fixed once and used throughout the Coxeter layer. It is therefore possible to read expressions such as

$$r_1 r_2, \qquad (r_1 r_2)^2, \qquad (r_1 r_2)^p$$

directly in the Lean source without repeatedly reopening the underlying definition.

2.3 The Coincident-Axis Check

The first sanity check is the case in which the two axes coincide.

```
@[simp]
theorem reflectionProduct_self_apply
    ...
    (axis : ReflectionAxis Geo)
    (P : Geo.Point) :
    reflectionProduct Geo axis axis P = P
```

This theorem is an immediate consequence of `lineReflect_involutive`. It confirms that the algebraic product layer is synchronized with the previously proved geometric relation

$$r_l^2 = 1.$$

The result is small, but architecturally important: the word layer is not an independent abstract group model. It is built directly on the actual Hilbert reflection transformations.

2.4 Powers of a Reflection Product

The module next introduces iteration of the two-generator product:

```
noncomputable def reflectionProductPow
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo)
    (n : Nat) :
    Equiv Geo.Point Geo.Point := ...
```

The associated normalization theorems are

```
reflectionProductPow_zero
reflectionProductPow_succ
reflectionProductPow_one
```

They provide the formal recursion rules needed later to reduce a numerical Coxeter exponent to an explicit word.

Schematically,

$$(r_1 r_2)^0 = 1,$$

and

$$(r_1 r_2)^{n+1} = (r_1 r_2)^n (r_1 r_2).$$

At this stage no claim is made about the order of $r_1 r_2$. The module only constructs the word. Its period must come from geometry of the pair of axes.

2.5 Satisfied Relation versus Exact Period

The first predicate is

```
def ReflectionPairRelation
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo)
    (p : Nat) : Prop :=
  reflectionProductPow Geo axis1 axis2 p =
    Equiv.refl Geo.Point
```

It asserts only the relation

$$(r_i r_j)^p = 1.$$

Accordingly, it says that the exponent p is *satisfied*; by itself it does not say that p is the least positive exponent. This distinction is exactly the one needed to match the "satisfied" versus "fulfilled" language emphasized by Coxeter–Moser.

The stronger predicate is part of the word layer:

```
def ReflectionPairExactPeriod
    (Geo : Geometry.Geo)
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo)
    (p : Nat) : Prop :=
  0 < p /\
  ReflectionPairRelation Geo axis1 axis2 p /\
  forall q : Nat,
    0 < q ->
    q < p ->
    Not (ReflectionPairRelation Geo axis1 axis2 q)
```

Thus

$$\texttt{ReflectionPairRelation}\, p$$

means that the displayed relation holds, whereas

$$\texttt{ReflectionPairExactPeriod}\, p$$

means that p is the actual positive period and can therefore be read as the Coxeter exponent of the pair.

The definition deliberately contains no geometric hypothesis on the axes. Geometry enters later, when a particular configuration proves both the relation and its minimality.

2.6 Symmetry under Reversing the Pair

The implementation convention for `reflectionProduct` remembers an order: `axis1` acts first and `axis2` second. The underlying Coxeter exponent, however, must not depend on which member of the pair is written first.

The module proves the pointwise conjugacy formula

```
theorem reflectionProductPow_swap_apply
    ...
    (n : Nat)
    (P : Geo.Point) :
    reflectionProductPow Geo axis2 axis1 n P =
      lineReflect Geo axis1
        (reflectionProductPow Geo axis1 axis2 n
          (lineReflect Geo axis1 P))
```

and packages its two consequences:

```
theorem reflectionPairRelation_symm
    ...
    (hRel :
      ReflectionPairRelation Geo axis1 axis2 p) :
    ReflectionPairRelation Geo axis2 axis1 p

theorem reflectionPairExactPeriod_symm
    ...
    (hExact :
      ReflectionPairExactPeriod Geo axis1 axis2 p) :
    ReflectionPairExactPeriod Geo axis2 axis1 p
```

This removes an implementation-order artifact from later public theorems. In particular, the equilateral $p = 3$ theorem may be stated with the geometrically natural public order (a, b), even when an intermediate word calculation is produced in the reverse order.

2.7 Why the Word Layer is Separate

It would have been possible to define the special relation for perpendicular axes directly in the geometric module. The separate word layer is more useful.

First, it fixes the algebraic language before any particular configuration is studied. Second, the same definitions can be reused for intersecting axes with other finite periods and for parallel axes, where no finite period is expected. Third, the final theorem can be stated in the same form as a Coxeter presentation rather than as an ad hoc pointwise identity.

The architecture is therefore

$$\begin{gathered}\texttt{Reflection.lean}\\ \downarrow\\ r_l \in \mathrm{Equiv}(\mathrm{Point})\\ \downarrow\\ \texttt{ReflectionWords.lean}\\ \downarrow\\ r_1 r_2,\ (r_1 r_2)^n\\ \Downarrow\\ \texttt{ReflectionPairRelation},\ \texttt{ReflectionPairExactPeriod}\\ \downarrow\\ \texttt{CoxeterRelations.lean}.\end{gathered}$$

2.8 Axiom Audit

The module introduces no new geometric axiom and no new geometric construction. It only packages previously constructed reflections as equivalences and composes them.

The noncomputability is inherited from `lineReflectEquiv`, whose underlying point map ultimately uses `Classical.choose` after geometric existence and uniqueness have already been proved.

The exact transitive axiom set of final theorems is obtained separately with `#print axioms`. No new foundational assumption is added by the word layer.

2.9 Conclusion

The module completes the passage from a single reflection generator to the language required for pairwise Coxeter relations:

$$\boxed{\begin{aligned} r_l^2 = 1 &\Longrightarrow r_1 r_2 \Longrightarrow (r_1 r_2)^p\\ &\Longrightarrow \texttt{ReflectionPairRelation}\\ &\Longrightarrow \texttt{ReflectionPairExactPeriod}\end{aligned}}$$

The word layer now distinguishes a relation that merely holds from the exact positive period of the product. What remains for each geometric configuration is to prove the relevant relation and then prove its minimality.

Chapter 3

Relations Between Two Reflection Generators

3.1 Purpose of the Chapter

This chapter records the first genuine relation between two distinct geometric reflection generators.

The target is the Coxeter case

$$p_{12} = 2.$$

Geometrically, this is the case in which the two reflecting axes are perpendicular. The development now proves both layers required for this statement:

$$(r_1 r_2)^2 = 1$$

and the minimality assertion that no positive exponent $q < 2$ gives the identity. Thus the final result is an exact Coxeter period, not merely a satisfied square relation.

The proof is not inserted as an abstract group identity. It is reconstructed from the Hilbert geometry of the two axes and from the previously proved metric behavior of line reflection.

Two Lean modules participate:

```
CGJteamLab/Coxeter/ReflectionIsometry.lean
CGJteamLab/Coxeter/CoxeterRelations.lean
```

The first establishes the metric behavior of a single reflection. The second packages perpendicular axes, proves the conjugation identity, and deduces the Coxeter relation.

3.2 Reflection as a Synthetic Isometry

The decisive theorem of `ReflectionIsometry.lean` is

```
theorem lineReflect_preserves_congruence
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis : ReflectionAxis Geo)
```

```
    (P Q : Geo.Point) :
    Geo.Congruent
      P Q
      (lineReflect Geo axis P)
      (lineReflect Geo axis Q)
```

Thus a line reflection preserves every segment congruence.

This is derived synthetically, without introducing coordinates or a numerical distance function. The proof separates several geometric configurations according to the perpendicular feet of the two points and uses the already established Hilbert congruence and right-angle theory.

Once segment congruence is available, the relation module extends the isometry package to the structural relations needed by conjugation:

```
lineReflect_preserves_between
lineReflect_preserves_collinear
lineReflect_preserves_noncollinear
lineReflect_preserves_angle
lineReflect_preserves_right_angle
```

The logical chain is

$$\begin{aligned} \text{segment congruence} &\Longrightarrow \text{betweenness and collinearity} \\ &\Longrightarrow \text{SSS} \\ &\Longrightarrow \text{angle congruence} \\ &\Longrightarrow \text{right-angle preservation.} \end{aligned}$$

This is precisely the transformation-theoretic package required to move a reflection configuration through another reflection.

3.3 Perpendicular Reflection Axes

Perpendicularity of the two reflection axes is represented by explicit witness data:

```
structure ReflectionAxesPerpendicular
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo) where
  O : Geo.Point
  U : Geo.Point
  V : Geo.Point
  hO1 : HilbertIncidence.OnLine O axis1.carrier
  hO2 : HilbertIncidence.OnLine O axis2.carrier
  hU1 : HilbertIncidence.OnLine U axis1.carrier
  hV2 : HilbertIncidence.OnLine V axis2.carrier
  hOU : Not (O = U)
  hOV : Not (O = V)
  hNonCol : Not (PrimCollinear Geo U O V)
  hRight : HilbertRightAngle Geo U O V
```

The certificate makes all hidden diagrammatic information explicit. The axes meet at O; the points U and V determine nondegenerate directions on the two carriers; and $\angle UOV$ is right.

Perpendicularity is then shown to be symmetric by `reflectionAxesPerpendicular_symm`.

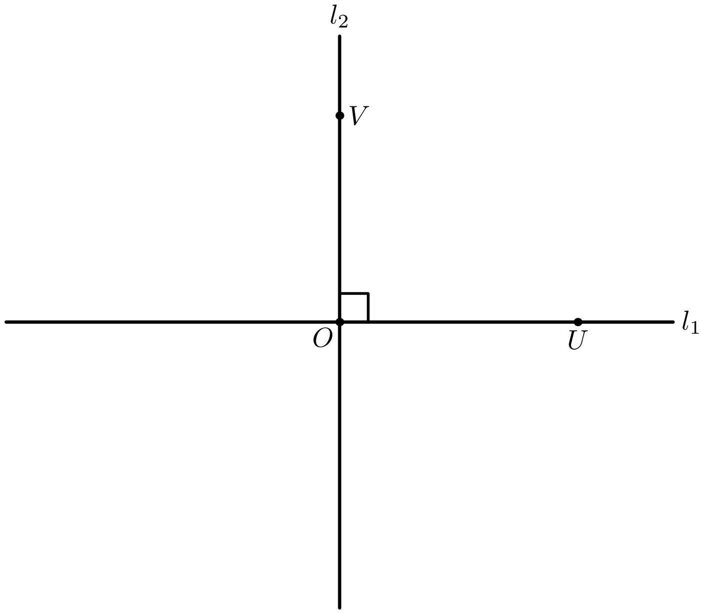


Figure 3.1: The synthetic $p = 2$ configuration. The reflection axes l_1 and l_2 meet at O at a right angle, with distinguished nondegenerate points $U \in l_1$ and $V \in l_2$.

3.4 Setwise Preservation of the Perpendicular Axis

The first geometric consequence is that reflection in one axis preserves the other perpendicular axis setwise.

The key theorems are

```
perpendicular_axes_second_off_first

perpendicular_axes_second_point_perpendicular_first

lineReflect_preserves_perpendicular_axis_second

lineReflect_preserves_perpendicular_axis_first
```

Their geometric content is simple but formally important.

Let $P \in l_2$ with $P \neq O$. Then $P \notin l_1$, otherwise the two carriers would share two distinct points and therefore coincide, contradicting the nondegenerate right-angle configuration.

Moreover OP is perpendicular to l_1. Therefore O is the perpendicular foot used in the reflection of P across l_1. The midpoint condition gives

$$P - O - r_1(P),$$

so $r_1(P)$ remains on the carrier l_2.

Hence

$$r_1(l_2) = l_2, \qquad r_2(l_1) = l_1.$$

This setwise invariance is necessary, but by itself it does not yet prove that the two reflections commute.

3.5 Transport of Midpoints and Perpendiculars

The conjugation argument requires more than preservation of the axis. A complete reflection witness consists of a perpendicular foot together with a midpoint condition.

The theorem

```
theorem lineReflect_preserves_midpoint
    ...
    (hMid : HilbertIsMidpoint Geo M A B) :
    HilbertIsMidpoint
      Geo
      (lineReflect Geo axis M)
      (lineReflect Geo axis A)
      (lineReflect Geo axis B)
```

uses preservation of strict betweenness and of segment congruence.

The second transport theorem is

```
theorem perpendicular_axes_transport_second_perpendicular
    ...
    (hPerp :
      PerpendicularToAxis Geo axis2 H P) :
    PerpendicularToAxis
      Geo
      axis2
      (lineReflect Geo axis1 H)
      (lineReflect Geo axis1 P)
```

Here the invariant carrier l_2 remains the target axis, while the foot, the off-axis point, and the right-angle witness are all transported through r_1.

This is the technical core of the conjugation proof.

3.6 Conjugation

The central pointwise theorem is

```
theorem perpendicular_axes_conjugation_pointwise
    ...
    (P : Geo.Point) :
    lineReflect Geo axis1
        (lineReflect Geo axis2
```

```
          (lineReflect Geo axis1 P)) =
      lineReflect Geo axis2 P
```

In mathematical notation,

$$r_1 r_2 r_1 = r_2.$$

The proof splits according to whether P lies on l_2.

If $P \in l_2$, then $r_1(P) \in l_2$, so r_2 fixes both relevant points and the conclusion reduces to $r_1^2 = 1$.

Assume now $P \notin l_2$. Reflect $r_1(P)$ in l_2, obtaining a perpendicular foot H and midpoint configuration

$$r_1(P) - H - r_2(r_1(P)).$$

Apply r_1 to the whole configuration. Preservation of the perpendicular data and of the midpoint gives

$$P - r_1(H) - r_1 r_2 r_1(P),$$

with $r_1(H) \in l_2$ and the required right-angle condition.

Therefore $r_1 r_2 r_1(P)$ satisfies the defining relation `IsLineReflection` for the original point P and the axis l_2.

But the reflected point in a fixed axis is unique. Hence

$$r_1 r_2 r_1(P) = r_2(P).$$

This is the exact point where the earlier theorem `line_reflection_unique` re-enters the Coxeter development.

3.7 Commutation

Applying the conjugation identity to $r_1(P)$ and using $r_1^2 = 1$ gives

```
theorem perpendicular_axes_reflections_commute_pointwise
    ...
    (P : Geo.Point) :
    lineReflect Geo axis1
        (lineReflect Geo axis2 P) =
      lineReflect Geo axis2
        (lineReflect Geo axis1 P)
```

Thus

$$r_1 r_2 = r_2 r_1$$

pointwise.

The equality is also packaged at the permutation level:

```
theorem perpendicular_axes_reflectionProduct_commute
    ...
    reflectionProduct Geo axis1 axis2 =
      reflectionProduct Geo axis2 axis1
```

This is the permutation-level form of the classical fact that reflections in perpendicular lines commute.

3.8 The Square of the Product

From the conjugation identity one obtains directly

```
theorem perpendicular_axes_reflectionProduct_square_apply
    ...
    (P : Geo.Point) :
    reflectionProduct Geo axis1 axis2
        (reflectionProduct Geo axis1 axis2 P) =
      P
```

The pointwise result is then lifted to an equality of equivalences:

```
theorem perpendicular_axes_reflectionProductPow_two
    ...
    reflectionProductPow Geo axis1 axis2 2 =
      Equiv.refl Geo.Point
```

The proof of this wrapper is intentionally thin. All substantive geometry has already been completed before this stage.

3.9 The Coxeter Relation and Exact Period $p_{12} = 2$

The first public word-layer wrapper is

```
theorem perpendicular_axes_coxeter_relation_two
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo)
    (hAxes :
      ReflectionAxesPerpendicular Geo axis1 axis2) :
    ReflectionPairRelation
      Geo
      axis1 axis2
      2
```

Unfolding `ReflectionPairRelation`, this is exactly

$$\boxed{(r_1 r_2)^2 = 1}.$$

This proves that the period divides 2. To identify the Coxeter exponent it remains to exclude period 1. The theorem is

```
theorem perpendicular_axes_reflections_exact_period_two
    (Geo : Geometry.Geo)
```

```
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (axis1 axis2 : ReflectionAxis Geo)
    (hAxes :
      ReflectionAxesPerpendicular Geo axis1 axis2) :
    ReflectionPairExactPeriod
      Geo
      axis1 axis2
      2
```

The minimality argument uses the distinguished point U from `ReflectionAxesPerpendicular`. Since $U \in l_1$, reflection in l_1 fixes U. Symmetry of perpendicularity and `perpendicular_axes_second_off_first` show that $U \notin l_2$, so reflection in l_2 moves U. Hence

$$(r_2 r_1)(U) \neq U,$$

and the product is not the identity. Since the square is already the identity, its least positive period is exactly 2.

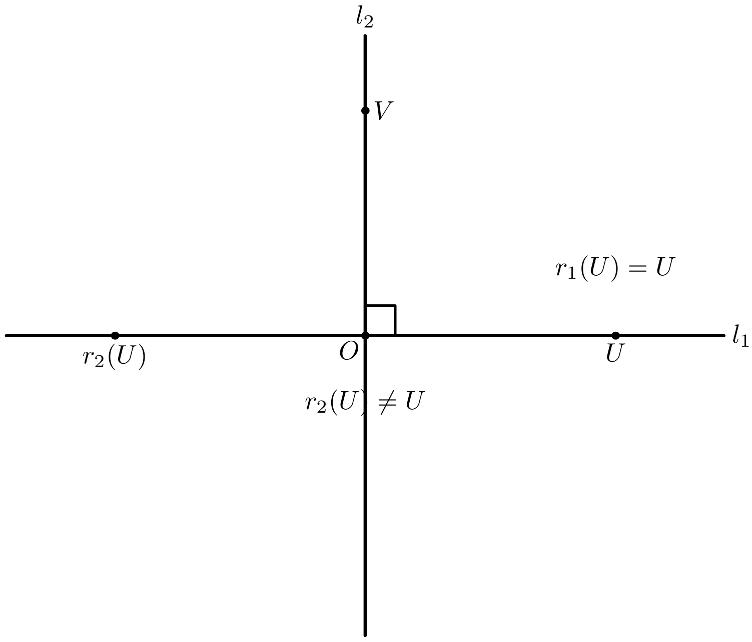


Figure 3.2: Witness for exact period 2. The point U is fixed by reflection in l_1, but lies off l_2 and is therefore moved by reflection in l_2. Hence the product is not the identity.

Thus the first nontrivial pairwise Coxeter exponent has now been derived in the strict minimal-period sense from the geometry of the mirrors:

$$\boxed{l_1 \perp l_2 \Longrightarrow p_{12} = 2.}$$

The theorem is stronger architecturally than an abstract presentation assumption. The generators themselves were reconstructed from Hilbert geometry, and the exponent 2 is derived from the geometric configuration of their axes.

3.10 Dependency Architecture

The completed proof may be summarized by the following DAG:

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence}\\
\Downarrow\\
\texttt{IsLineReflection} \longrightarrow \texttt{lineReflect} \longrightarrow r_l^2 = 1\\
\Downarrow\\
\texttt{lineReflect\_preserves\_congruence}\\
\Downarrow\\
\text{betweenness, collinearity, angles, right angles}\\
\Downarrow\\
\texttt{ReflectionAxesPerpendicular}\\
\Downarrow\\
r_1(l_2) = l_2, \quad r_2(l_1) = l_1\\
\Downarrow\\
\text{midpoint and perpendicular transport}\\
\Downarrow\\
r_1 r_2 r_1 = r_2\\
\Downarrow\\
r_1 r_2 = r_2 r_1\\
\Downarrow\\
(r_1 r_2)^2 = 1\\
\Downarrow\\
\texttt{ReflectionPairRelation}\ 2\\
\Downarrow\\
\text{the product moves } U\\
\Downarrow\\
\boxed{\texttt{ReflectionPairExactPeriod}\ 2}.
\end{gathered}
$$

This gives a complete passage from synthetic geometry to a nontrivial relation among several Coxeter generators.

3.11 Axiom Audit

The two modules documented here introduce no new declared geometric axiom and no `sorry`.

The public relation theorem is stated with only `HilbertIncidence` and `HilbertCongruence` instances. The transitive axiom footprint of the final theorem is recorded with

```
#print axioms Geometry.perpendicular_axes_reflections_exact_period_two
```

The $p = 2$ relation and its minimality are proved inside the established Hilbert infrastructure without adding a Coxeter-specific geometric axiom.

3.12 Conclusion

The completed chapter establishes the chain

$$\boxed{\text{perpendicular mirrors} \Longrightarrow r_1 r_2 r_1 = r_2 \Longrightarrow r_1 r_2 = r_2 r_1 \Longrightarrow (r_1 r_2)^2 = 1.}$$

In the language prepared by `ReflectionWords.lean`, the chapter now ends with both levels:

$$\texttt{ReflectionPairRelation}\ 2, \qquad \boxed{\texttt{ReflectionPairExactPeriod}\ 2}.$$

The second statement is the one that justifies reading the Coxeter label literally as $p_{12} = 2$.

Chapter 4

The Coxeter Relation $p = 3$: Equilateral Median Reflections

4.1 Purpose of the Chapter

The perpendicular case $p = 2$ is the first nontrivial pairwise Coxeter relation, but it is still exceptional: the two reflections commute. The present chapter records the first genuinely noncommutative case obtained in the synthetic Hilbert development.

The target is

$$p_{ab} = 3.$$

Rather than introducing a numerical angle of 60°, the construction uses an equilateral triangle and its three median reflection axes. If a, b, c denote these axes, the synthetic geometry proves the two carrier transports

$$r_b(a) = c, \qquad r_a(b) = c.$$

The general conjugation theorem then yields

$$r_b r_a r_b = r_c, \qquad r_a r_b r_a = r_c,$$

and therefore the braid relation

$$\boxed{r_a r_b r_a = r_b r_a r_b}.$$

Using only involutivity of the two reflections,

$$r_a^2 = r_b^2 = 1,$$

this gives

$$\boxed{(r_a r_b)^3 = 1}.$$

The relation is packaged in the word layer, where the positive period is proved to be exactly 3.

The corresponding Lean source is

```
CGJteamLab/Coxeter/CoxeterRelations.lean
```

4.2 Why the $p = 3$ Step is Structurally Different

For perpendicular axes the decisive geometric fact was setwise preservation: reflection in one axis preserves the other axis. This produced

$$r_1 r_2 r_1 = r_2,$$

hence commutation and the square relation.

The $p = 3$ case cannot be obtained by repeating that argument. Reflection in one median axis of an equilateral triangle does not preserve a second median axis. It sends that axis to the third one.

Thus the correct geometric pattern is no longer

$$r_a(b) = b,$$

but rather

$$r_a(b) = c.$$

This is the first place where the Coxeter layer needs a genuine transport theory for reflection axes. The new proof architecture is

$$\begin{gathered}\text{reflection of points}\\ \Downarrow\\ \text{reflection of a carrier line}\\ \Downarrow\\ \text{conjugation of line reflections}\\ \Downarrow\\ \text{braid relation.}\end{gathered}$$

The passage from $p = 2$ to $p = 3$ is therefore not merely a change of exponent. It introduces the mechanism by which one reflection transports another reflection generator to a different generator.

4.3 No Numerical Angle is Introduced

The construction is deliberately synthetic.

No distance function, coordinate system, scalar product, trigonometric function, or numerical angle measure is introduced. In particular, the proof does not define the angle between a and b to be $60°$, and it does not use the analytic identity

$$2\cos(\pi/3) = 1.$$

Instead, the order-three relation is extracted from a finite geometric configuration whose symmetries can be reconstructed using the already available Hilbert notions:

- segment congruence,
- midpoint,
- collinearity and betweenness,

- right angle and perpendicularity,
- line reflection,
- preservation of midpoint data by reflection,
- uniqueness of a midpoint.

The equilateral triangle is therefore not being used as an analytic model of an angle. It is used as a synthetic incidence–congruence configuration that forces the required permutation of median axes.

4.4 The Equilateral Configuration

Let A, B, C be noncollinear points satisfying the three apex forms of the equilateral side condition:

$$AB \cong AC, \qquad BA \cong BC, \qquad CB \cong CA.$$

The Lean hypotheses are oriented in exactly this way because the median-axis construction is invoked once from each vertex.

Let

$$M_A = \mathrm{mid}(B, C), \qquad M_B = \mathrm{mid}(A, C), \qquad M_C = \mathrm{mid}(B, A).$$

The three reflection axes are then constructed as the median carriers

$$a = AM_A, \qquad b = BM_B, \qquad c = CM_C.$$

The choice

$$M_C = \mathrm{mid}(B, A)$$

rather than initially orienting the midpoint as $\mathrm{mid}(A, B)$ is only a formal endpoint convention. The public midpoint relation is symmetric, and the existing theorem

```
MidpointSymmetry
```

is used when the opposite orientation is required.

4.5 The Median Axis of an Isosceles Triangle

The geometric constructor underlying all three axes is

```
isosceles_midpoint_axis_swaps_base
```

Its input consists of an isosceles triangle with apex A, base BC, and a midpoint M of BC. The theorem constructs a reflection axis through A and M and proves that the associated line reflection fixes the apex and midpoint while swapping the base endpoints:

$$r(A) = A, \qquad r(M) = M, \qquad r(B) = C, \qquad r(C) = B.$$

Schematically, the theorem has the form

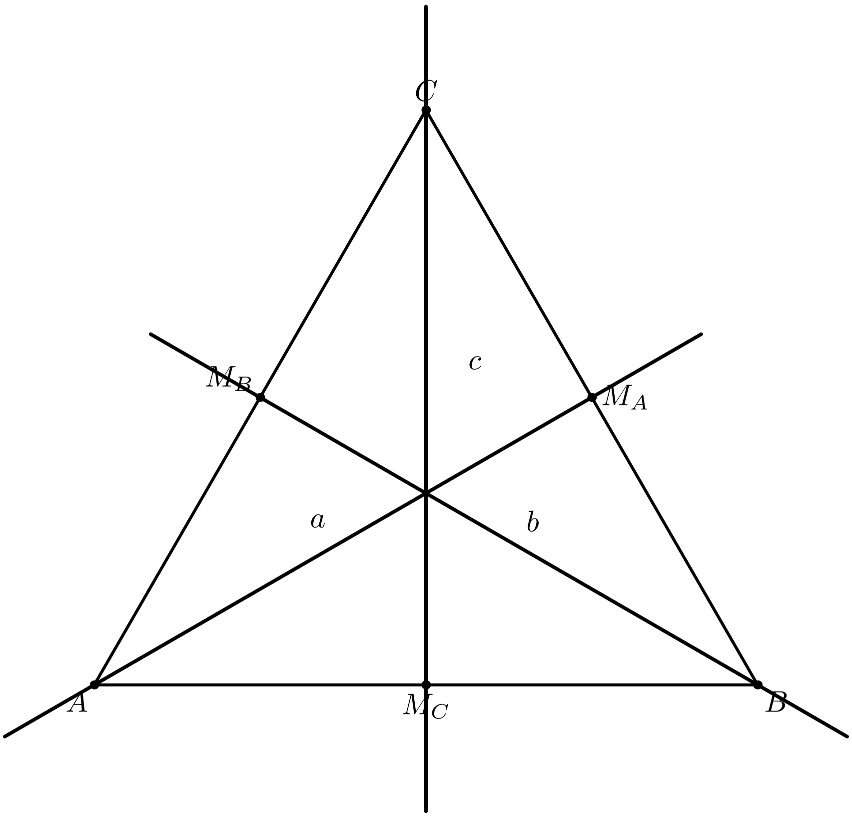


Figure 4.1: The equilateral configuration for $p = 3$. The three median carriers a, b, c are the reflection axes used throughout the chapter.

```
theorem isosceles_midpoint_axis_swaps_base
    ...
    (hABC : Not (Collinear Geo A B C))
    (hABAC : Geo.Congruent A B A C)
    (hMid : HilbertIsMidpoint Geo M B C) :
    Exists fun axis : ReflectionAxis Geo =>
      ...
```

The proof itself depends on the synthetic right-angle theorem for the median of an isosceles triangle:

```
isosceles_midpoint_right_angle
```

Thus the median is promoted to an actual reflection axis without assuming a global symmetry of the triangle as a primitive transformation.

Applying the theorem at the three vertices gives:

axis	fixed points	swapped vertices
a	A, M_A	$B \leftrightarrow C$
b	B, M_B	$A \leftrightarrow C$
c	C, M_C	$B \leftrightarrow A.$

This finite table is the geometric engine of the $p = 3$ proof.

4.6 Transporting an Entire Carrier Line

Pointwise reflection of the vertices is not yet enough for conjugation. Conjugation requires knowledge of the image of the whole carrier of a reflection axis.

The relation

```
def ReflectionMapsLine
    ...
    (mirror : ReflectionAxis Geo)
    (source target : Geo.Line) : Prop :=
  forall P : Geo.Point,
    HilbertIncidence.OnLine P source <->
      HilbertIncidence.OnLine
        (lineReflect Geo mirror P)
        target
```

encodes exact setwise transport of one line onto another.

The meaning of

$$\mathrm{ReflectionMapsLine}(b, a, c)$$

is

$$P \in a \quad \Longleftrightarrow \quad r_b(P) \in c.$$

This is stronger than merely proving that two selected points of a land on c. The selected points are instead used as a criterion from which the full line transport is derived.

4.7 The Two-Point Criterion

The helper

```
reflectionMapsLine_of_two_points
```

is the bridge from point geometry to carrier geometry.

Suppose X, Y are distinct points on the source carrier and their reflected images both lie on the target carrier. Incidence uniqueness then forces the reflected source line to coincide with the target line, giving

$$\mathrm{ReflectionMapsLine}\,.$$

In the $p = 3$ proof the source points are chosen to be the two defining points of a median carrier:

$$A, M_A \in a$$

or

$$B, M_B \in b.$$

Thus the main problem becomes the precise determination of the images of the midpoints M_A and M_B.

4.8 Midpoint Transport Under Reflection

The already established reflection-isometry layer provides

```
lineReflect_preserves_midpoint
```

which transports a midpoint configuration through any line reflection.

From

$$M_A = \mathrm{mid}(B, C)$$

and reflection in b, we obtain

$$r_b(M_A) = \mathrm{mid}\big(r_b(B), r_b(C)\big).$$

But the median reflection b fixes B and swaps C with A:

$$r_b(B) = B, \qquad r_b(C) = A.$$

Hence

$$r_b(M_A)$$

is a midpoint of BA.

The designated point M_C is also a midpoint of BA. Therefore midpoint uniqueness gives

$$\boxed{r_b(M_A) = M_C}.$$

The local uniqueness lemma used here is

```
hilbert_midpoint_unique_local
```

with statement of the form

```
theorem hilbert_midpoint_unique_local
    ...
    (hM : HilbertIsMidpoint Geo M A B)
    (hN : HilbertIsMidpoint Geo N A B) :
    M = N
```

This is one of the key synthetic moves in the chapter. No coordinates are needed to calculate the reflected midpoint. Its image is identified by the universal geometric property of being the midpoint of the reflected endpoints.

4.9 First Carrier Transport: $r_b(a) = c$

The first genuine $p = 3$ theorem is

```
equilateral_median_axis_b_maps_a_to_c
```

The proof uses the two defining points of a:

$$A,\ M_A.$$

Reflection in b gives

$$r_b(A) = C$$

and, by midpoint transport and uniqueness,

$$r_b(M_A) = M_C.$$

Both C and M_C lie on the carrier c. Since $A \neq M_A$, the two-point criterion yields

$$\boxed{r_b(a) = c}.$$

In Lean the conclusion is represented as

```
ReflectionMapsLine Geo b a.carrier c.carrier
```

rather than as an equality between primitive line-image objects.

This theorem shows that reflection in one Coxeter generator can transport the carrier of a second generator to the carrier of a third.

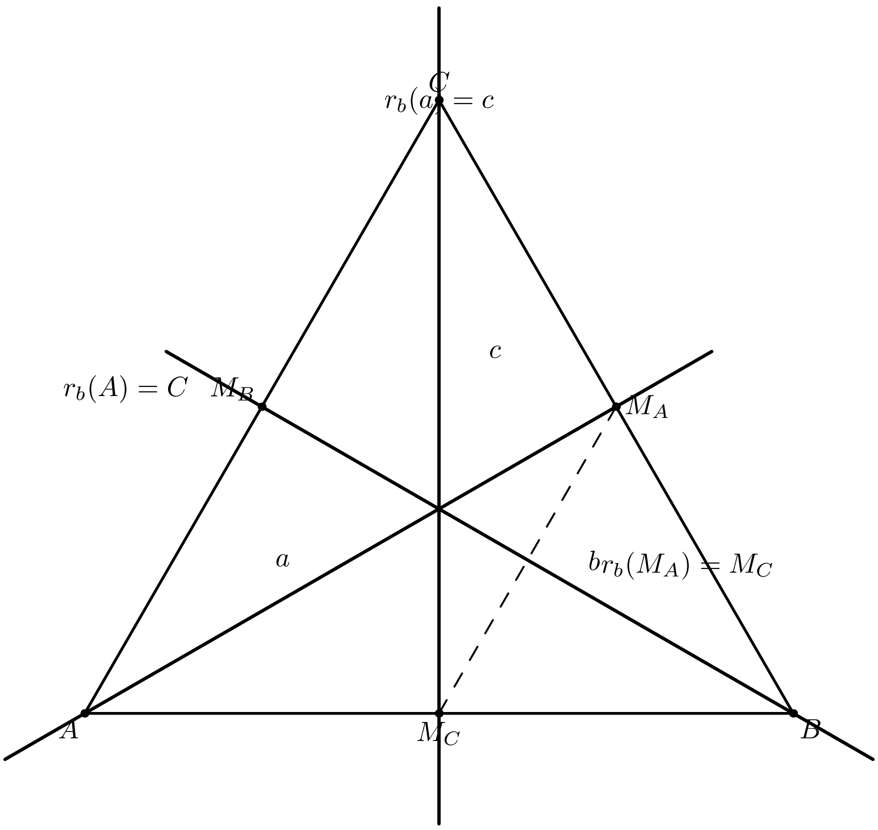


Figure 4.2: Carrier transport under r_b. Since $r_b(A) = C$ and $r_b(M_A) = M_C$, the whole median carrier $a = AM_A$ is transported to $c = CM_C$.

4.10 General Conjugation by Carrier Transport

The carrier transport theorem feeds directly into the general conjugation principle

```
lineReflect_conjugation_pointwise
```

whose geometric content is: if reflection in m sends the carrier of s onto the carrier of t, then

$$r_m r_s r_m = r_t.$$

More precisely, from

$$r_m(s) = t$$

the theorem proves pointwise

$$r_m(r_s(r_m(P))) = r_t(P).$$

Applying this with

$$m = b, \qquad s = a, \qquad t = c$$

gives

$$\boxed{r_b r_a r_b = r_c}.$$

The corresponding Lean theorem is

```
equilateral_median_reflection_conjugation_bab_eq_c
```

This isolates the group-theoretic consequence of the first carrier transport before any braid identity is used.

4.11 Second Carrier Transport on the Same Axes

To derive the braid relation, it is essential that both conjugation identities refer to the *same* existentially chosen axes a, b, c. Constructing a second unrelated triple of axes would leave an additional identification problem.

The synchronization theorem

```
equilateral_median_axes_two_transports
```

therefore constructs one triple a, b, c and proves simultaneously

$$r_b(a) = c$$

and

$$r_a(b) = c.$$

The second transport is proved in exact analogy with the first.

Since a fixes A and swaps B with C,

$$r_a(A) = A, \qquad r_a(C) = B.$$

Transporting

$$M_B = \mathrm{mid}(A, C)$$

through r_a shows that $r_a(M_B)$ is a midpoint of AB.

The designated midpoint M_C was initially oriented as a midpoint of BA. The theorem

`MidpointSymmetry`

converts this to the required midpoint of AB, after which uniqueness gives

$$r_a(M_B) = M_C.$$

Together with

$$r_a(B) = C$$

the two-point criterion yields

$$\boxed{r_a(b) = c}.$$

4.12 The Braid Relation

The two carrier transports now give two conjugation identities on the same axes:

$$r_b r_a r_b = r_c,$$

$$r_a r_b r_a = r_c.$$

Comparing them immediately gives

$$\boxed{r_a r_b r_a = r_b r_a r_b.}$$

The Lean theorem is

```
equilateral_median_reflections_braid
```

and its final proof step is simply the transitivity of equality through the common reflection r_c.

This is the braid relation associated with the two generators.

It is also the first relation in the Coxeter layer in which the two generators do not collapse to a commutative pair. For $p = 2$, the geometry forced

$$r_1 r_2 = r_2 r_1.$$

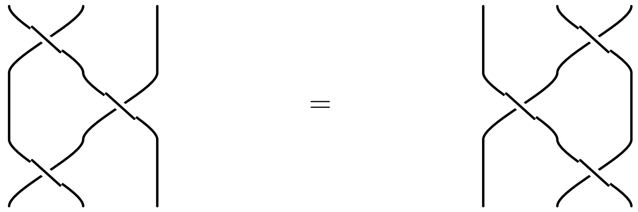
For $p = 3$, the correct reduced relation is instead

$$r_a r_b r_a = r_b r_a r_b.$$

This is the standard braid form associated with the rank-two Coxeter system of type A_2, equivalently $I_2(3)$.

There is a useful distinction here. The same word identity is one of the standard Artin relations for the braid group, but braid generators are not involutions. The Coxeter group is obtained by imposing, in addition, the relations $s_i^2 = 1$. Thus the diagram below illustrates the *shape* of the braid relation independently of the synthetic reflection geometry that proves it synthetically. For the classical theory, see the nLab entry *braid group*.[1]

The two figures in this section have complementary roles. Figure 4.3 is an abstract strand diagram for the Artin relation, while Figure 4.4 records the synthetic geometric reason that the corresponding reflection words coincide.



$b_1b_2b_1$ $b_2b_1b_2$

Figure 4.3: The Artin braid relation on three strands. The two diagrams are isotopic, giving $b_1b_2b_1 = b_2b_1b_2$. The Coxeter braid relation has the same word form, while the simple reflections additionally satisfy $s_i^2 = 1$.

4.13 From the Braid Relation to Order Three

Once the braid relation is available, no further geometry is needed.

The final theorem of the present chain is

```
equilateral_median_reflections_product_order_three
```

and proves pointwise

$$(r_a r_b)^3(P) = P.$$

Expanded literally, the left-hand side is

$$r_a r_b r_a r_b r_a r_b(P).$$

Using

$$r_a r_b r_a = r_b r_a r_b$$

on the first three factors gives

$$r_b r_a r_b r_b r_a r_b(P).$$

Now involutivity cancels the adjacent equal reflections:

$$r_b r_a \underbrace{r_b r_b}_{1} r_a r_b(P) = r_b \underbrace{r_a r_a}_{1} r_b(P) = \underbrace{r_b r_b}_{1}(P) = P.$$

Thus

$$\boxed{(r_a r_b)^3 = 1}.$$

[1] https://ncatlab.org/nlab/show/braid+group, accessed 4 September 2026.

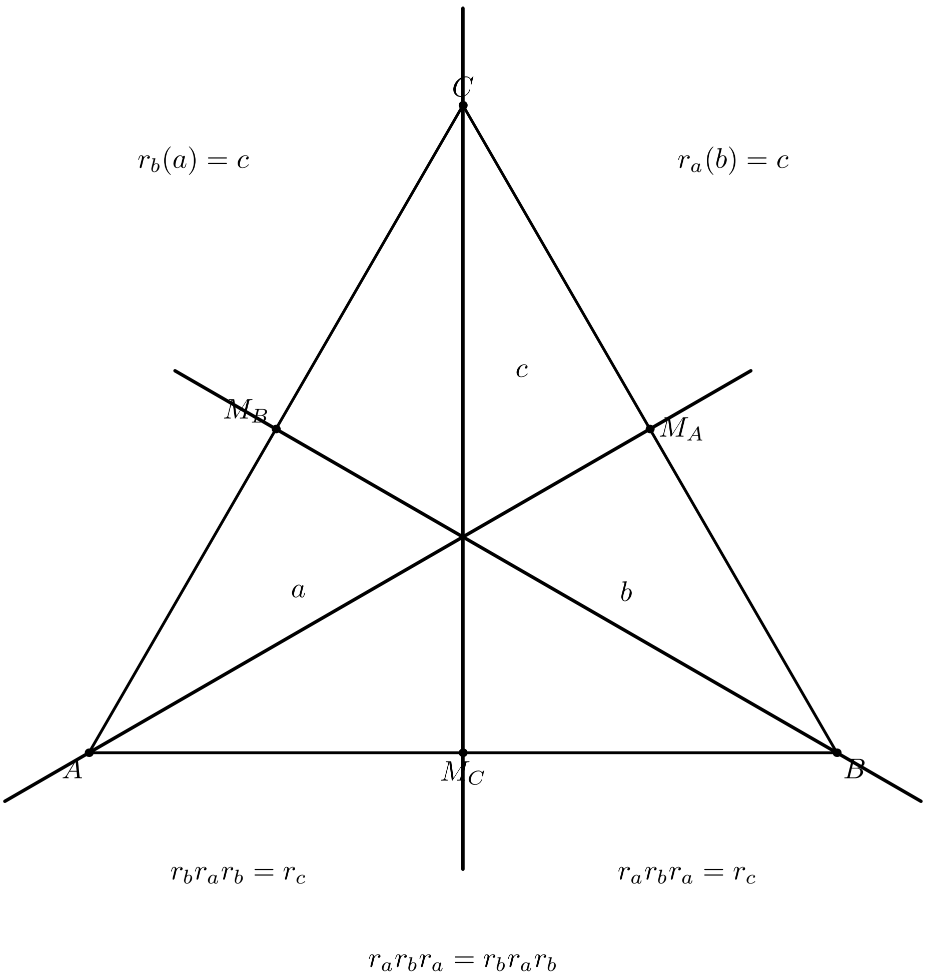


Figure 4.4: Synthetic geometric origin of the braid relation. Both conjugations produce the same third reflection r_c, hence $r_a r_b r_a = r_b r_a r_b$.

The proof is intentionally separated from the geometric part. All geometric work has already been completed by the carrier transports and conjugation theorems. The order-three result is then a short word calculation using only the braid relation and involutivity.

4.14 Relation Between the Two Forms of the $p = 3$ Relation

For involutions $a^2 = b^2 = 1$, the two familiar forms

$$aba = bab$$

and

$$(ab)^3 = 1$$

encode the same rank-two Coxeter relation.

They arise here in the geometrically natural order:

$$\text{carrier transport} \Longrightarrow \text{conjugation} \Longrightarrow aba = bab \Longrightarrow (ab)^3 = 1.$$

This order is important for the formal architecture. The braid identity is not guessed algebraically and then checked. It is forced by the fact that both a and b, under conjugation by the other, are transported to the same third geometric reflection c.

The third median axis is therefore not auxiliary decoration. It is the geometric witness that explains why the braid relation holds.

4.15 Dependency Architecture

The $p = 3$ chain may be summarized as follows:

$$
\begin{gathered}
\texttt{lineReflect\_preserves\_midpoint} \\
\Downarrow \\
\texttt{isosceles\_midpoint\_axis\_swaps\_base} \\
\Downarrow \\
a, b, c \text{ median reflection axes} \\
\Downarrow \\
\texttt{hilbert\_midpoint\_unique\_local} \\
\Downarrow \\
r_b(M_A) = M_C, \quad r_a(M_B) = M_C \\
\Downarrow \\
\texttt{reflectionMapsLine\_of\_two\_points} \\
\Downarrow \\
r_b(a) = c, \quad r_a(b) = c \\
\Downarrow \\
\texttt{lineReflect\_conjugation\_pointwise} \\
\Downarrow \\
r_b r_a r_b = r_c, \quad r_a r_b r_a = r_c \\
\Downarrow \\
\boxed{r_a r_b r_a = r_b r_a r_b} \\
\Downarrow \\
r_a^2 = r_b^2 = 1 \\
\Downarrow \\
(r_a r_b)^3 = 1 \\
\Downarrow \\
\texttt{ReflectionPairRelation}\ 3 \\
\Downarrow \\
\text{the product moves } B \\
\Downarrow \\
\texttt{ReflectionPairExactPeriod}\ 3.
\end{gathered}
$$

The crucial new middle layer, absent from the original $p = 2$ proof, is

$$\text{point transport} \longrightarrow \text{carrier transport} \longrightarrow \text{reflection conjugation.}$$

4.16 Axiom Audit

No new Coxeter-specific geometric axiom is introduced in the $p = 3$ construction.

The argument is built over the already established Hilbert incidence and congruence infrastructure together with the derived reflection machinery. The local theorem statements use the same typeclass assumptions as the preceding reflection relation layer.

The transitive axiom footprint of the final public theorem is obtained by Lean audit, for example with

```
#print axioms
  Geometry.equilateral_median_reflections_exact_period_three
```

The local claim is that no new geometric axiom or `sorry` is introduced in order to obtain the $p = 3$ relation.

4.17 Word-Layer Packaging and Exact Period

The pointwise theorem

$$(r_a r_b)^3(P) = P$$

has now been exported to the word layer. The first public wrapper is

```
theorem equilateral_median_reflections_coxeter_relation_three
    ...
    Exists fun a : ReflectionAxis Geo =>
      Exists fun b : ReflectionAxis Geo =>
        Exists fun c : ReflectionAxis Geo =>
          ...
          ReflectionPairRelation Geo a b 3
```

Thus the formalization proves

$$\texttt{ReflectionPairRelation}\ 3.$$

This still says only that the period divides 3. Exactness is established by the stronger theorem

```
theorem equilateral_median_reflections_exact_period_three
    ...
    Exists fun a : ReflectionAxis Geo =>
      Exists fun b : ReflectionAxis Geo =>
        Exists fun c : ReflectionAxis Geo =>
          ...
          ReflectionPairExactPeriod Geo a b 3
```

The minimality proof is again geometric. The vertex B lies on the median axis b, so $r_b(B) = B$. The same vertex cannot lie on a: otherwise A, B, C would all lie on the carrier a, contradicting the noncollinearity hypothesis. Therefore

$$r_a(B) \neq B.$$

With the implementation convention for `reflectionProduct`, the intermediate product

$$\texttt{reflectionProduct Geo}\ b\ a$$

acts as $r_a r_b$, so it moves B. Hence exponent 1 is impossible.

For a positive exponent below 3, the only remaining possibility is 2. If both the square and cube of the product were the identity, the recursive power law would force the product itself to be the identity, contradicting the moved point B. Thus exponent 2 is impossible as well.

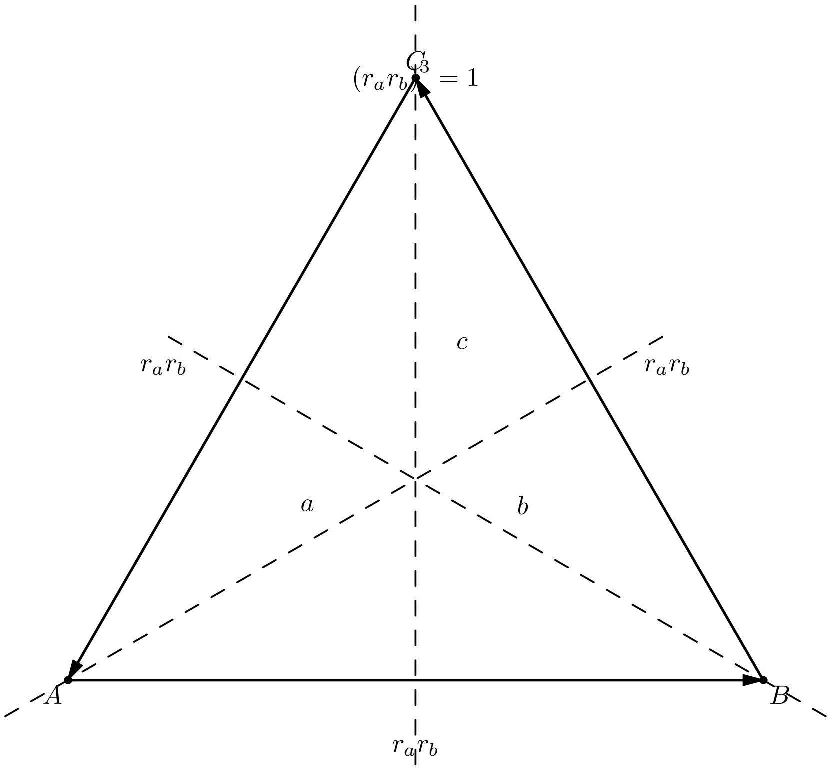
The intermediate calculation naturally appears in the order (b, a). The general theorem

`reflectionPairExactPeriod_symm`

then removes this implementation-order artifact and gives the public statement in the order (a, b).

The implication chain is therefore

$$
\boxed{
\begin{aligned}
\text{synthetic geometry} &\Longrightarrow aba = bab \\
&\Longrightarrow (ab)^3 = 1 \\
&\Longrightarrow \texttt{ReflectionPairRelation}\ 3 \\
&\Longrightarrow \texttt{ReflectionPairExactPeriod}\ 3.
\end{aligned}
}
$$



$$B \mapsto C, \quad B \neq C$$

Figure 4.5: The vertex cycle underlying exact period 3. The product is nontrivial on the vertices, while its cube is the identity.

4.18 Geometric Significance

The $p = 2$ case may be viewed as a special orthogonality theorem, whereas the $p = 3$ case exhibits a more general mechanism. The geometry need not preserve each reflection generator individually: it may permute a finite family of reflection axes, and this permutation can be converted into conjugation identities among the corresponding generators.

For the equilateral configuration,

$$a, \ b, \ c$$

form exactly such a finite family. Consequently the Coxeter exponent is recovered from a purely synthetic configuration, without first introducing numerical angle measure. This carrier-transport mechanism is the structural content of the chapter beyond the particular exponent 3.

4.19 Conclusion

The completed $p = 3$ chain is

$$\boxed{\begin{array}{c} \text{equilateral triangle} \\ \Downarrow \\ \text{three median reflection axes } a, b, c \\ \Downarrow \\ r_b(a) = c, \quad r_a(b) = c \\ \Downarrow \\ r_b r_a r_b = r_c, \quad r_a r_b r_a = r_c \\ \Downarrow \\ r_a r_b r_a = r_b r_a r_b \\ \Downarrow \\ (r_a r_b)^3 = 1 \\ \Downarrow \\ \boxed{\texttt{ReflectionPairExactPeriod}\,3}. \end{array}}$$

The proof is synthetic throughout. The exponent 3 is not imported from coordinates, trigonometry, or a numerical 60° angle. It is forced by the reflection geometry of the equilateral configuration.

The word-layer bridge and the minimality proof are now complete:

$$\boxed{\texttt{ReflectionPairExactPeriod}\,3}.$$

Chapter 5

General Intersecting Axes and Finite Period

5.1 Purpose of the Chapter

The preceding chapters established the exact Coxeter exponents $p = 2$ and $p = 3$ by two special synthetic configurations. This chapter gives a uniform finite-period mechanism for a product of two Hilbert line reflections.

Two Lean modules are involved:

```
CGJteamLab/Coxeter/CoxeterRelationsGeneral.lean
CGJteamLab/Coxeter/CoxeterRelationsGeneralExistence.lean
```

The first module is the structural layer. It develops carrier transport, telescoping of products of reflections, closure of finite carrier chains, and minimality of the resulting period. The second module identifies a synthetic geometric object from which the required carrier chain can be generated and isolates the remaining existence principle.

The result is deliberately not formulated through a numerical angle π/p. The formal route is instead

$$\boxed{\begin{array}{c}\text{finite radial polygon data}\\ \Downarrow\\ \text{reflection transport of successive radial carriers}\\ \Downarrow\\ \text{telescoping carrier chain}\\ \Downarrow\\ \text{first carrier return at } p\\ \Downarrow\\ \texttt{ReflectionPairExactPeriod}\ p.\end{array}}$$

For $p = 2$, no additional polygon-existence principle is required: a nondegenerate Hilbert right angle produces two perpendicular reflection axes, and the exact-period-two theorem from the previous relation chapter applies.

For $p \geq 3$, the formal theorem is conditional on the existence of an oriented equilateral radial

p-polygon. This condition is made explicit in the Lean interface and is the present geometric existence boundary of the general finite-period development.

5.2 Why the General Step is Not an Angle-Measure Theorem

The usual Euclidean description of two intersecting reflection axes says that their product is a rotation through twice the angle between the mirrors. A finite Coxeter exponent p is then associated with an angle of the form π/p.

That is not the proof language used here. No numerical angle, trigonometric identity, cosine, coordinate model, or metric formula is introduced. The general mechanism is stated entirely through synthetic incidence, order, segment congruence, line reflection, and finite recurrence.

The geometric replacement for "angle π/p" is a finite family of radial points

$$V_0, V_1, \dots, V_p$$

about a center O, together with congruence and separation conditions that make the successive radial carriers behave as a coherent finite fan.

Thus the exponent is read from a first-return phenomenon:

$$\text{first return of the radial carrier} \quad \Longrightarrow \quad \text{exact period of the reflection product.}$$

This is the same conceptual reversal used throughout the supplement: Coxeter–Moser provide the generators-and-relations vocabulary, while the formal proof reconstructs the relation from synthetic geometry.

5.3 The General Carrier Telescope

The structural core of `CoxeterRelationsGeneral.lean` is independent of any particular polygon.

Let

$$a_0, a_1, a_2, \dots$$

be reflection axes and assume that reflection in the middle axis transports the preceding carrier to the following carrier:

$$r_{a_{n+1}}(\text{carrier}(a_n)) = \text{carrier}(a_{n+2}).$$

In Lean this local geometric condition is represented by `ReflectionMapsLine`. The conjugation theorem converts it into an identity of adjacent reflection products:

$$r_{a_{n+1}} r_{a_n} = r_{a_{n+2}} r_{a_{n+1}}.$$

Consequently all consecutive products in the chain agree with the initial product. If

$$T = r_{a_1} r_{a_0},$$

then the powers of T telescope:

$$T^n = r_{a_n} r_{a_0}.$$

The point of this identity is that a relation for T becomes a geometric statement about the n-th carrier.

If the p-th carrier is the initial carrier, then

$$r_{a_p} = r_{a_0}$$

as canonical line reflections, and hence

$$T^p = r_{a_p} r_{a_0} = r_{a_0}^2 = 1.$$

More importantly, if no positive carrier return occurs before p, then no smaller positive power of T can be the identity. The theorem

```
coxeter_general_exact_period_of_minimal_closed_carrier_line
```

therefore turns a first carrier return into an exact Coxeter period.

This separates the argument into two layers:

$$\boxed{\text{carrier recurrence and first return} \Longrightarrow \text{exact period.}}$$

The remaining task is geometric: construct a finite carrier family with the required recurrence and minimal closure.

5.4 Canonical Radial Axes

Fix a center O and a sequence of points

$$V_0, V_1, \ldots, V_p$$

with $O \neq V_n$. The general module associates to each V_n the canonical reflection axis through O and V_n.

The intended carrier chain is therefore

$$OV_0,\ OV_1,\ OV_2, \ldots, OV_p.$$

To prove the Coxeter recurrence it is enough to show, for each local window

$$V_n, \quad V_{n+1}, \quad V_{n+2},$$

that reflection in the middle radial axis OV_{n+1} transports the carrier OV_n onto OV_{n+2}.

The first version of this mechanism was obtained from an isosceles triangle. If

$$OV_n \cong OV_{n+2},$$

and a midpoint M_n of V_nV_{n+2} lies on the middle radial axis, then reflection in that axis exchanges the two outer vertices and transports the whole preceding radial carrier to the following one.

This is the local geometric engine behind the general carrier recurrence.

5.5 The Diameter Obstruction and the $p = 4$ Check

An important correction appeared when the local midpoint construction was tested against the square.

An earlier radial-fan interface required

$$M_n \neq O$$

for every midpoint M_n of the two-step chord

$$V_n V_{n+2}.$$

That condition is too strong. In the square case, the two-step chord is a diameter. Its midpoint is exactly the center:

$$M_n = O.$$

Thus a global `midpoint_off_center` hypothesis would exclude the perfectly legitimate Coxeter exponent $p = 4$. This is not merely a Lean technicality; it exposes a genuine geometric branch that the first nondegenerate isosceles argument did not cover.

The corrected local theorem therefore splits into two cases.

5.5.1 The non-diameter branch

If

$$M \neq O,$$

equal radii imply that O is not between the two outer vertices. The resulting noncollinearity gives the ordinary isosceles-midpoint reflection argument, and the middle radial axis transports the previous carrier to the next one.

5.5.2 The diameter branch

If

$$M = O,$$

then the outer vertices lie on one diameter through O. The equal-chord condition at the middle vertex makes the middle radial axis perpendicular to that diameter.

The existing perpendicular-axis reflection theorem then shows that reflection in the middle radial axis preserves the diameter carrier setwise. Since the previous and next radial carriers are the same diameter line, this is exactly the required neighbor transport.

The interface records the setwise preservation step as

```
coxeter_general_reflection_maps_perpendicular_carrier_self
```

and packages the perpendicular certificate in

```
coxeter_general_perpendicular_radial_axes_of_center_midpoint
```

before deriving

```
coxeter_general_diameter_neighbor_transport
```

and finally the uniform local statement

```
coxeter_general_neighbor_transport_with_diameter_case.
```

The resulting theorem contains no global exclusion of diameter chords.

5.6 The Raw Polygon Object

The existence module introduces

```
structure HilbertFiniteEquilateralRadialPolygon
```

for $p \geq 3$.

Its fields express the following data.

1. The exponent satisfies
$$3 \leq p.$$
2. The center is distinct from the initial radial point:
$$O \neq V_0.$$
3. All radial segments have one common congruence class:
$$OV_n \cong OV_0 \qquad (n \leq p).$$
4. Consecutive polygon sides have one common congruence class, locally expressed as
$$V_nV_{n+1} \cong V_{n+1}V_{n+2} \qquad (n+1 < p).$$
5. The chain closes:
$$V_p = V_0.$$
6. No positive index $q < p$ gives an early radial return to the initial carrier:
$$O,\ V_q,\ V_0 \quad \text{are not collinear.}$$

The last field is already a minimality condition. It is what later rules out every smaller positive Coxeter relation.

The object is intentionally weaker in vocabulary than a numerically described regular polygon. It uses only points, segment congruence, closure, and radial nonreturn.

5.7 Why Orientation Data are Needed

Equal radii and equal consecutive side lengths alone do not force the local sequence to progress coherently around the center. A finite point sequence could backtrack or zigzag while preserving those congruences.

The structure

```
structure HilbertFiniteOrientedEquilateralRadialPolygon
```

therefore adds a Hilbert-order condition.

For every local triple

$$V_n,\ V_{n+1},\ V_{n+2},$$

there exists a radial line through

$$O, \quad V_{n+1}$$

such that the two outer vertices V_n and V_{n+2} lie on opposite sides of that line.

This is the field

```
local_radial_separation
```

and it supplies the synthetic orientation that would otherwise be hidden in a numerical angle parameter.

The condition has a second decisive role. Since the segment V_nV_{n+2} crosses the separating radial line, there is an intersection point M. The common-radius condition makes O equidistant from the outer vertices, while the equal-side condition makes V_{n+1} equidistant from them. The already established Hilbert theorem on an equidistant line bisecting a segment then proves that M is the midpoint of the two-step chord.

Thus midpoint data are reconstructed locally from the raw polygon conditions; they are no longer part of the input structure.

5.8 The Carrier Chain from Polygon Data

The theorem

```
coxeter_general_carrier_chain_with_diameter_case
```

performs the local construction uniformly.

For every n with

$$n + 1 < p,$$

it obtains the radial separator from the orientation field, extracts the intersection point M with the two-step chord, proves that M is its Hilbert midpoint, and applies the two-branch neighbor-transport theorem.

The polygon-level wrapper

```
coxeter_general_carrier_chain_of_oriented_equilateral_polygon
```

therefore produces the entire recurrence

$$r_{OV_{n+1}}(OV_n) = OV_{n+2}$$

in the carrier sense needed by the general telescope.

No midpoint family is supplied by the user, and no `midpoint_off_center` hypothesis remains.

This is a significant simplification of the formal interface:

$$\boxed{\text{raw congruence} + \text{Hilbert separation} \Longrightarrow \text{local midpoint} \Longrightarrow \text{carrier transport.}}$$

5.9 Closure and Exact Period

The theorem

```
coxeter_general_exact_period_of_oriented_equilateral_polygon
```

is the main unconditional polygon-to-Coxeter result of the existence module.

Given an actual object

```
HilbertFiniteOrientedEquilateralRadialPolygon Geo p O V
```

it constructs the canonical axes through

$$O, V_0 \qquad \text{and} \qquad O, V_1$$

and proves

$$\boxed{\texttt{ReflectionPairExactPeriod}\, p.}$$

The proof now has a transparent geometric decomposition:

$$\begin{array}{c} \text{local radial separation and congruence} \\ \Downarrow \\ \text{carrier recurrence} \\ \Downarrow \\ V_p = V_0 \\ \Downarrow \\ \text{carrier}(a_p) = \text{carrier}(a_0) \\ \Downarrow \\ (r_{a_0} r_{a_1})^p = 1, \end{array}$$

while

$$\begin{gathered}0 < q < p \\ \Downarrow \\ O, V_q, V_0 \text{ not collinear} \\ \Downarrow \\ \text{carrier}(a_q) \neq \text{carrier}(a_0) \\ \Downarrow \\ (r_{a_0} r_{a_1})^q \neq 1.\end{gathered}$$

Hence closure gives the relation and radial nonreturn gives its minimality.

5.10 Equal Half-Turn Division as an Existence Principle

An oriented radial polygon gives one explicit existence interface for arbitrary finite period. A more elementary interface is

```
class HilbertFiniteEqualAngleHalfTurnDivisionExistence
```

Its content is entirely synthetic: for every natural number p with $3 \leq p$, there exists an oriented half-pencil and a finite chain of p congruent angular steps beginning on one boundary ray and ending exactly on the opposite boundary ray. No numerical angle measure, cosine, trigonometry, or coordinate model is part of this interface.

The proved bridge is

$$\boxed{\begin{gathered}\text{terminal equal-angle chain of length } p \\ \Downarrow \\ \text{normalized equilateral semicircle division} \\ \Downarrow \\ \text{oriented radial polygon data} \\ \Downarrow \\ \texttt{ReflectionPairExactPeriod}\ p.\end{gathered}}$$

The corresponding uniform theorem is

```
coxeter_general_exact_period_exists_of_equal_halfturn_division_ge_two
```

and proves that, under `HilbertFiniteEqualAngleHalfTurnDivisionExistence`, every $p \geq 2$ occurs as the exact period of a product of two Hilbert line reflections.

For the Coxeter-theoretic interpretation this class may be read as the synthetic existence principle that a p-fold equal division of a half-turn can be chosen.

The class `HilbertFiniteOrientedEquilateralRadialPolygonExistence` provides a valid intermediate interface, and the corresponding polygon-to-period theorems factor through it.

5.11 Period Two Requires No Additional Existence Principle

The theorem

```
coxeter_general_exact_period_two_exists
```

closes the lower endpoint of the finite range directly in the ordinary Hilbert congruence layer.

The proof chooses two distinct points on a line, extends the segment, invokes the established theorem giving a nondegenerate right angle, takes the two carriers of that right angle as reflection axes, and applies

```
perpendicular_axes_reflections_exact_period_two.
```

Hence

$$\exists a\, b, \qquad \texttt{ReflectionPairExactPeriod}\,\texttt{Geo}\ a\ b\ 2$$

without the polygon-existence typeclass.

The uniform rank-two theorem is

```
coxeter_general_exact_period_exists_of_equal_halfturn_division_ge_two
```

and states that, under the equal-half-turn-division principle, every finite exponent $p \geq 2$ occurs as the exact period of a product of two Hilbert line reflections.

The logical split is therefore

$$\boxed{\begin{array}{lcl} p = 2 & : & \text{ordinary Hilbert incidence/order/congruence,} \\ p \geq 3 & : & \text{ordinary Hilbert layer} + \text{ equal finite division of a half-turn.} \end{array}}$$

5.12 Dependency Architecture

The dependency chain is

$$\boxed{\begin{array}{c} \texttt{Reflection.lean} \\ \Downarrow \\ \texttt{ReflectionIsometry.lean} \\ \Downarrow \\ \texttt{CoxeterRelations.lean} \\ \Downarrow \\ \texttt{CoxeterRelationsGeneral.lean} \\ \Downarrow \\ \texttt{CoxeterRelationsGeneralExistence.lean}. \end{array}}$$

Conceptually, the last two modules divide the general finite-period problem as

`CoxeterRelationsGeneral`
carrier algebra and telescoping

$\Downarrow$

`CoxeterRelationsGeneralExistence`
synthetic finite-angle data, normalization, and existence interfaces.

The reflection and telescoping layers do not depend on how equal finite division of a half-turn is obtained. The existence principle is therefore logically separated from the rank-two Coxeter mechanism.

5.13 Axiom Audit

The two general finite-period modules introduce no coordinate geometry, no numerical angle measure, and no trigonometric machinery.

The structural theorem in `CoxeterRelationsGeneral.lean` is derived from the existing reflection and Hilbert infrastructure. The existence module contains a chain of explicit geometric interfaces, culminating in `HilbertFiniteEqualAngleHalfTurnDivisionExistence` for the uniform rank-two theorem.

This distinction must be preserved in any axiom audit:

1. the implication

$$\text{oriented radial polygon of period } p \Longrightarrow \texttt{ReflectionPairExactPeriod}\, p$$

   is proved;

2. the equal-angle-chain normalization and semicircle-division bridges are proved;

3. the period-two existence theorem is proved from the established Hilbert layer;

4. uniform existence for all $p \geq 3$ is isolated in the explicit class `HilbertFiniteEqualAngleHalfTurnDivisionExistence`.

The appropriate audit commands are the theorem names, for example

```
#print axioms Geometry.coxeter_general_exact_period_two_exists
#print axioms Geometry.coxeter_general_exact_period_of_oriented_equilateral_polygon
#print axioms Geometry.
    coxeter_general_exact_period_exists_of_equal_halfturn_division_ge_two
```

with the final line interpreted together with its explicit equal-division typeclass assumption.

5.14 Conclusion

The general finite-period chapter establishes the following architecture:

$$\boxed{\begin{array}{c}\text{Hilbert reflection generators}\\ \Downarrow\\ \text{carrier transport}\\ \Downarrow\\ T^n = r_{a_n} r_{a_0}\\ \Downarrow\\ \text{first carrier return}\\ \Downarrow\\ \text{exact Coxeter period}\\ \Downarrow\\ \text{oriented equilateral radial } p\text{-polygon}\\ \Downarrow\\ \texttt{ReflectionPairExactPeriod}\ p.\end{array}}$$

The square case forced an essential repair: diameter chords cannot be excluded, because for $p = 4$ the midpoint of the two-step chord is the center. The final local transport theorem therefore handles both non-diameter and diameter configurations.

For $p = 2$, existence is already obtained from a Hilbert right angle. For $p \geq 3$, the Coxeter relation and its exactness are reduced to the explicit existence of an equal finite division of a half-turn. If that geometric principle is accepted, the rank-two Coxeter problem is complete.

Chapter 6

The Three-Dimensional Coxeter System A_3: Plane Reflections, Tetrahedral Realization, and S_4

6.1 Purpose and Status

This chapter records the first completed genuinely spatial Coxeter realization in CGJteamLab. The target is the Coxeter system of type A_3, realized by reflections in three planes of a Hilbert three-dimensional space.

The production modules are

```
CGJteamLab/Coxeter/PlaneReflection.lean
CGJteamLab/Coxeter/PlaneReflectionIsometry.lean
CGJteamLab/Coxeter/CoxeterRelations3D.lean
CGJteamLab/Coxeter/CoxeterRelations3DExistence.lean
CGJteamLab/Coxeter/CoxeterRelations3DS4.lean
CGJteamLab/Coxeter/CoxeterA3Wyler.lean
```

The development now has four logically distinct layers.

First, assuming a tetrahedral reflection frame, the three plane reflections are shown to satisfy the global Coxeter relations of type A_3. Second, the frame itself is constructed synthetically from the current three-dimensional Hilbert geometry. Third, the generated subgroup is identified with the full symmetric group on the four tetrahedral vertices. Fourth, the completed frame is re-read through the Wyler flat calculus: mirror intersections are computed as flats, their joins reconstruct the mirrors, their carriers are identified with exact common fixed loci, and explicit planar slices recover the local Coxeter exponents $3, 3, 2$.

Thus the final state is no longer conditional. The production theorem

```
coxeter_A3_tetrahedral_frame_exists
```

proves that the geometric frame needed by the Coxeter-relation layer actually exists, while

```
CoxeterA3Vertex.generatedGroup_isomorphic_S4
```

packages the final group-theoretic identification

$$\langle r_1, r_2, r_3 \rangle \cong S_4.$$

The entire construction remains synthetic. It uses no coordinates, vectors, inner products, Gram matrices, determinants, numerical angle measure, cosine, or trigonometry.

6.2 Why the A_3 Step is Structurally New

The rank-two chapters concern reflections in lines of a plane. The A_3 problem is qualitatively different for two reasons.

First, the generators are now reflections in planes rather than reflections in lines. A reflected point therefore has to be constructed from a perpendicular to a plane and a midpoint condition.

Second, the three generators have to be realized simultaneously by one spatial configuration. The problem is not merely to realize three pairwise periods separately. One needs a single noncoplanar tetrahedral frame in which

$$(r_1 r_2)^3 = 1, \qquad (r_2 r_3)^3 = 1, \qquad (r_1 r_3)^2 = 1$$

hold at the same time.

This is the first concrete instance of the simultaneous higher-rank program announced in the front matter.

6.3 The Three-Dimensional Hilbert Architecture

The ambient three-dimensional development deliberately does not identify the whole space with a planar Hilbert geometry. Spatial incidence, order, and congruence are represented by dedicated interfaces such as

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

while planar theorems are recovered only inside an explicitly chosen plane.

The bridge is

```
PlaneGeo Geo pi
```

for a spatial plane π. Inside this slice one has access to the established two-dimensional Hilbert and Euclidean theorem library.

This distinction is fundamental. In particular, the proof does not add a global ambient instance

```
HilbertCongruence Geo
```

merely in order to reuse planar lemmas. Whenever a problem is genuinely coplanar, it is moved into a specific `PlaneGeo`. Whenever two configurations lie in different planes, an explicitly spatial transport theorem is proved.

The result is a controlled interface between two-dimensional and three-dimensional reasoning.

6.4 Reflection in a Plane

The first new geometric primitive of the chapter is not postulated as a primitive transformation. As in the line-reflection chapter, it is reconstructed from existence and uniqueness data.

A perpendicular from a point P to a plane π through a foot F is represented by

```
def PerpendicularToPlaneThrough
    (pi : S.Plane)
    (F P : Geo.Point) : Prop :=
  exists l : Geo.Line,
    H.OnLine P l /\
    HilbertLinePerpendicularPlaneAt Geo l pi F
```

The reflected point P' across π is characterized by the familiar midpoint geometry: either $P \in \pi$ and $P' = P$, or the perpendicular foot F is the midpoint of PP'.

This leads to the relation

```
IsPlaneReflection
```

and then to the selected point transformation

```
planeReflect
```

together with its permutation packaging

```
planeReflectEquiv
```

The same pattern as in line reflection therefore survives in three dimensions:

$$\text{geometric relation} \Longrightarrow \text{existence} \Longrightarrow \text{uniqueness} \Longrightarrow \text{point transformation} \Longrightarrow \text{permutation.}$$

The reflection is involutive, and its fixed points are exactly the points of the reflecting plane.

6.5 Plane Reflection as a Spatial Isometry

The decisive metric theorem is

```
planeReflect_preserves_congruence
```

which states that reflection in a plane preserves segment congruence.

The proof is synthetic. The relevant point pair and perpendicular data are placed in a suitable planar slice, where the established line-reflection isometry theorem can be reused. The resulting congruence statement is then transported back to the ambient spatial geometry.

This module is important conceptually because it avoids rebuilding every metric property of reflection from scratch in three dimensions. The spatial proof is reduced to a two-dimensional reflection theorem inside the correct section plane.

6.6 The Tetrahedral Coxeter Frame

The relation module is organized around the structure

```
CoxeterA3TetrahedralFrame
```

which stores four points

$$A,\ B,\ C,\ D$$

with the noncoplanarity condition

$$\neg\,\mathrm{HilbertCoplanar4}(A, B, C, D),$$

together with three reflecting planes π_1, π_2, π_3.

The intended action is:

$$r_1 :\ A \leftrightarrow B, \qquad C, D \text{ fixed},$$

$$r_2 :\ B \leftrightarrow C, \qquad A, D \text{ fixed},$$

$$r_3 :\ C \leftrightarrow D, \qquad A, B \text{ fixed}.$$

Each mirror is specified by a midpoint and perpendicular condition for the edge which it bisects, together with the requirement that the other two vertices lie in the plane.

This is the geometric realization of the three adjacent transpositions

$$(AB), \qquad (BC), \qquad (CD).$$

6.7 Vertex Action and the Coxeter Words

Once the three plane reflections are available, their action on the four distinguished vertices can be computed directly.

For the adjacent pairs, the product cyclically permutes three vertices. Therefore the cubes act trivially on all four vertices:

```
r12_cube_eq_refl
r23_cube_eq_refl
```

For the nonadjacent pair, the generators act on disjoint vertex pairs and the square is the identity:

```
r13_sq_eq_refl
```

At this point these are not yet merely formal group-presentation statements. They are consequences of the explicitly constructed geometric action of three plane reflections on one tetrahedral frame.

6.8 Four-Point Rigidity

The key globalization principle is a spatial rigidity theorem:

> a spatial isometry which fixes four noncoplanar points is the identity.

This is the three-dimensional analogue of using sufficiently many fixed points to determine a rigid motion.

The proof is entirely synthetic. Distances from an arbitrary point to the four fixed vertices are preserved by the isometry. The development proves the required uniqueness statement for a point determined by these four distance constraints. Noncoplanarity is exactly the condition which prevents the fourth constraint from being redundant.

This theorem converts equalities verified on the tetrahedral vertices into global equalities of transformations of the entire point space.

6.9 Global Coxeter Relations of Type A_3

Applying four-point rigidity to the previously computed vertex action gives the production theorem

```
coxeter_A3_relations
```

whose geometric content is

$$r_1^2 = r_2^2 = r_3^2 = 1,$$

$$(r_1 r_2)^3 = 1, \qquad (r_2 r_3)^3 = 1, \qquad (r_1 r_3)^2 = 1.$$

The corresponding braid and commutation forms are also proved explicitly:

```
r1_r2_braid
r2_r3_braid
r1_r3_commute
```

that is,

$$r_1 r_2 r_1 = r_2 r_1 r_2,$$

$$r_2 r_3 r_2 = r_3 r_2 r_3,$$

and

$$r_1 r_3 = r_3 r_1.$$

These are global equalities of the actual plane-reflection transformations, not merely identities on the four vertices.

From the Artin-group viewpoint, the three displayed identities are exactly the type-A_3 braid pattern: adjacent generators satisfy the length-three braid relation, while the distant generators commute. The Artin group of type A_3 is the four-strand braid group Br_4; adding the involutivity relations $r_i^2 = 1$ gives the Coxeter group $W(A_3)$. The strand picture in Figure 4.3 therefore applies twice here, to (r_1, r_2) and (r_2, r_3), while r_1 and r_3 are the distant commuting pair. The later theorem `generatedGroupMulEquivS4` identifies the resulting geometric Coxeter group with S_4, so the formal development realizes the familiar quotient from the Artin presentation geometrically rather than merely postulating it.

6.10 The Remaining Geometric Problem

At this stage the A_3 relation theorem was still conditional: it assumed the existence of a `CoxeterA3TetrahedralFrame`.

The second part of the development removes precisely this assumption.

The final existence chain is

$$\boxed{\begin{array}{c} \text{three-square corner} \\ \Downarrow \\ \text{regular tetrahedron} \\ \Downarrow \\ \text{three edge-bisecting mirror planes} \\ \Downarrow \\ \texttt{CoxeterA3TetrahedralFrame}. \end{array}}$$

The crucial design decision was to avoid a direct "intersection of three spheres" construction. Such a route would require a much larger locus-and-sphere infrastructure.

Instead, the regular tetrahedron is extracted from a configuration built from three squares.

6.11 The Regular Tetrahedron Interface

The intermediate structure is

```
HilbertRegularTetrahedron3D
```

with vertices A, B, C, D, a noncoplanarity certificate, and five congruence conditions to the reference edge AB:

$$AC \cong AB, \qquad AD \cong AB, \qquad BC \cong AB,$$

$$BD \cong AB, \qquad CD \cong AB.$$

No numerical length function is introduced. Regularity is expressed only through the existing segment-congruence relation.

6.12 From a Regular Tetrahedron to a Mirror Plane

The central existence lemma is

```
hilbert_edge_mirror_plane_exists
```

Suppose F is the midpoint of AB, while C and D are each equidistant from A and B:

$$CA \cong CB, \qquad DA \cong DB.$$

The midpoint geometry gives

$$FC \perp AB, \qquad FD \perp AB.$$

A preliminary lemma

```
hilbert_noncoplanar4_midpoint_opposite_noncollinear
```

shows that F, C, D are not collinear when A, B, C, D are noncoplanar. Thus F, C, D determine a plane.

Euclid XI.4 now supplies the crucial spatial step: a line perpendicular at one point to two distinct intersecting lines of a plane is perpendicular to the plane itself. Hence

$$AB \perp \pi$$

at F.

The proof also shows $A \notin \pi$. Otherwise the uniqueness of the perpendicular foot, in the form supplied by the XI.12 layer, would force the endpoint A to coincide with the midpoint F.

The plane π is therefore precisely the reflection plane which swaps A and B and fixes C, D.

The same construction applied to the edges BC and CD produces the other two mirrors.

This is packaged as

```
HilbertRegularTetrahedron3D.toCoxeterA3TetrahedralFrame
```

6.13 Why a Direct Tetrahedron Construction Was Avoided

A direct construction of a regular tetrahedron would naturally ask for a point simultaneously equidistant from three given vertices. In analytic geometry this is immediate from coordinates or sphere intersections.

That is not the proof language used here.

The synthetic development instead constructs a square base, erects an equal normal edge, builds two vertical square faces, and then takes the four alternating vertices.

The classical visual model is the familiar fact that four alternating vertices of a cube form a regular tetrahedron. The Lean proof, however, does not assume a cube object and does not use coordinates of cube vertices. Every required congruence is proved from the local square and right-angle geometry.

6.14 Spatial SAS and Right-Angle Transport

The three-square construction exposed an important interface issue.

The two right triangles which must be compared often lie in different planes. A planar SAS theorem cannot simply be applied globally to the ambient space.

The development therefore introduces the spatial bridge

```
hilbert_space_sas_third_side
```

which allows two triangles in different planes to be compared through the spatial congruence interface.

For configurations already known to be coplanar, the chapter uses

```
hilbert_space_coplanar_right_angles_congruent
hilbert_space_coplanar_right_triangles_hypotenuse
```

and for genuinely cross-plane comparison it proves the spatial forms

```
hilbert_space_adjacent_angles_congruent
hilbert_space_all_right_angles_congruent
hilbert_space_right_triangles_hypotenuse
```

The middle theorem is the spatial analogue of Hilbert's theorem that all right angles are congruent. The last theorem is the workhorse used later: two right triangles in different planes with corresponding legs congruent have congruent hypotenuses.

Several auxiliary same-ray and angle-transport lemmas support this construction, including

```
hilbert_space_sameRay_of_between
hilbert_space_sameRay_ray_eq
hilbert_space_angle_eq_of_sameRay_first
hilbert_space_angle_eq_of_sameRay_second
hilbert_space_noncollinear_of_sameRays
hilbert_space_between_transport_sameRays
```

These results make the cross-plane angle calculus explicit rather than hiding it behind an ambient planar instance.

6.15 Constructing the Three-Square Corner

The base square is obtained from the planar Euclidean layer. The helper

```
hilbert_square_complete_of_equal_perpendicular
```

packages the familiar construction: two equal perpendicular sides determine a square.

The spatial lemma

```
hilbert_space_equal_normal_segment_exists
```

then constructs, from a point A in a plane π and a nondegenerate reference segment AB, a point E on a normal through A such that

$$E \notin \pi, \qquad AE \cong AB.$$

The combined scaffold theorem

```
hilbert_space_square_equal_normal_scaffold_exists
```

therefore supplies

$$ABCD \subset \pi$$

as a square together with a normal edge AE of the same length.

To package a square living in a named spatial plane, the development introduces

```
IsSquareInPlane
```

which hides the dependent `PlanePoint` witnesses and allows the existing planar `IsSquare` predicate to be reused cleanly.

The theorem

```
hilbert_space_vertical_square_exists
```

then builds a vertical square $ABFE$ on the base edge AB and normal edge AE. Repeating the construction on AD gives a second vertical square $ADHE$.

The complete configuration is stored in

```
HilbertThreeSquareCorner3D
```

and its existence is proved by

```
hilbert_three_square_corner_exists
```

The three square faces are therefore

$$ABCD, \qquad ABFE, \qquad ADHE.$$

6.16 Transporting Perpendicularity to the Opposite Vertical Edges

The normality of AE to the base plane is not enough by itself. Later right-triangle comparisons require the opposite vertical edges BF and DH also to be perpendicular to the base.

The bridge

```
hilbert_space_parallel_carriers_of_planeGeo_parallel
```

converts planar parallelism of square edges into ambient parallelism of their carrier lines.

Euclid XI.8 then transports perpendicularity to the plane along a parallel line.

The result is

```
HilbertThreeSquareCorner3D.vertical_edges_perpendicular_to_base
```

which yields carrier lines satisfying

$$BF \perp \pi \qquad \text{and} \qquad DH \perp \pi.$$

This is the synthetic version of the elementary fact that the vertical edges of a rectangular box are parallel normals to the base plane.

6.17 The Alternating Vertices

The regular tetrahedron is extracted from the four alternating vertices

$$A, \ C, \ F, \ H.$$

The main congruence theorem is

```
HilbertThreeSquareCorner3D.alternating_edges_congruent
```

which proves

$$AF \cong AC, \qquad AH \cong AC, \qquad CF \cong AC,$$

$$CH \cong AC, \qquad FH \cong AC.$$

Together with the reference edge AC, this gives all six equal edges.

The first four comparisons use the spatial right-triangle theorem.

For example, $\triangle ABC$ and $\triangle ABF$ are right triangles with corresponding legs supplied by the base and front squares. Hence

$$AC \cong AF.$$

The same mechanism yields AH, CF, and CH.

6.18 The Subtle Edge FH

The edge FH deserves separate attention because no top square is constructed.

The proof instead derives a new perpendicularity.

In the base square,

$$AB \perp AD.$$

Since AE is perpendicular to the base plane,

$$AE \perp AB.$$

Thus AB is perpendicular to two intersecting lines AD and AE of the left-face plane $ADHE$. Euclid XI.4 gives

$$AB \perp \text{plane}(ADHE).$$

The front square gives

$$AB \parallel FE.$$

Euclid XI.8 transports the plane-perpendicularity to FE, and therefore

$$FE \perp EH.$$

Hence $\triangle FEH$ is right. Its two legs are congruent to the two legs of $\triangle ABC$, so

$$FH \cong AC.$$

This argument is important for the architecture: the regular tetrahedron is obtained without ever constructing the fourth, upper face of a cube.

6.19 A Planar Obstruction to Four Pairwise Equal Points

Equal edge lengths do not yet prove that A, C, F, H form a genuine tetrahedron. They could in principle be coplanar.

The key planar theorem is

```
hilbert_no_four_pairwise_equal_points
```

It states that in a single Hilbert plane there cannot be four points whose six pairwise segments are all congruent to one fixed nondegenerate segment.

The proof is entirely synthetic.

Take A, B, C, D with all six pairwise distances equal.

If C and D lie on the same side of the base line AB, Euclid I.7 forces

$$C = D,$$

because both points have the same prescribed distances from A and B. But CD is assumed congruent to the nondegenerate base segment, so $C \neq D$. Contradiction.

If C and D lie on opposite sides of AB, the segment CD meets the base line. The intersection point lies on one of the two base rays, producing a strict angle inclusion. At the same time SSS and same-ray transport identify the larger and smaller angles by congruence. The angle-order relation can therefore be transported to obtain

$$\alpha < \alpha,$$

contradicting

```
hilbert_angleLess_irrefl
```

This theorem replaces what a coordinate proof might express through dimension or determinant arguments.

6.20 Noncoplanarity of the Alternating Vertices

The theorem

```
HilbertThreeSquareCorner3D.alternating_noncoplanar
```

uses the planar obstruction.

Assume that A, C, F, H lie in a plane ρ. Passing to

```
PlaneGeo Geo rho
```

transports all previously proved segment congruences into one planar geometry. One obtains exactly the forbidden configuration of four points with all six pairwise segments equal.

Therefore

$$\neg\,\mathrm{HilbertCoplanar4}(A, C, F, H).$$

This is a characteristic example of the project method: a genuinely three-dimensional fact is proved by reducing the hypothetical degenerate case to a contradiction inside a two-dimensional slice.

6.21 Existence of the Regular Tetrahedron

Once the five congruences and noncoplanarity are available, the definition

```
HilbertThreeSquareCorner3D.toRegularTetrahedron
```

takes

$$A_{\mathrm{tet}} = A, \qquad B_{\mathrm{tet}} = C, \qquad C_{\mathrm{tet}} = F, \qquad D_{\mathrm{tet}} = H$$

and packages the resulting data as a `HilbertRegularTetrahedron3D`.

The existence theorem is

```
hilbert_regular_tetrahedron3D_exists
```

This is a significant foundational point. The regular tetrahedron is not assumed as an additional spatial axiom; it is constructed from the existing Hilbert and Euclidean spatial interface.

6.22 Existence of the A_3 Reflection Frame

The final existence theorem is

```
coxeter_A3_tetrahedral_frame_exists
```

Its proof is now conceptually short:

$$\texttt{hilbert\_three\_square\_corner\_exists}$$

$$\Downarrow$$

$$\texttt{hilbert\_regular\_tetrahedron3D\_exists}$$

$$\Downarrow$$

$$\texttt{HilbertRegularTetrahedron3D.toCoxeterA3TetrahedralFrame}$$

$$\Downarrow$$

$$\texttt{CoxeterA3TetrahedralFrame}.$$

The geometric difficulty lies entirely in the previous sections. The final assembly merely packages the completed construction.

Combined with `CoxeterRelations3D.lean`, this gives an unconditional synthetic realization of the Coxeter relations of type A_3.

6.23 Axiomatic Status of the Geometric Existence Chain

The final audit command is

```
#print axioms Geometry.coxeter_A3_tetrahedral_frame_exists
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Quot.sound]
```

No new three-dimensional existence axiom appears.

In particular, the development does not postulate

- existence of a regular tetrahedron;
- existence of an A_3 Coxeter frame;
- coordinates on the spatial point set;
- vectors or an inner product;
- a numerical distance function;
- a numerical angle function;
- a Gram matrix or positivity condition.

The use of `Classical.choice` comes from selecting witnesses after geometric existence has already been proved.

6.24 Production Merge and Dependency Audit

The existence proof was developed incrementally through a sequence of isolated test files. Before promotion to production, the complete chain was merged into a validation module which imported no test module.

This check exposed one hidden dependency on the previously proved spatial right-triangle theorem. After the missing earlier bridge layer was included, the merged file compiled independently.

The final production file was then cleaned structurally, rebuilt, and audited again.

The resulting endpoint

```
Geometry.coxeter_A3_tetrahedral_frame_exists
```

therefore represents a self-contained production theorem rather than a result accidentally depending on experimental test modules.

6.25 From the Tetrahedral Frame to the Symmetric Group

The geometric $3D/A_3$ stage leaves one natural algebraic question. The three plane reflections act on the four tetrahedral vertices as

$$(AB), \qquad (BC), \qquad (CD).$$

The expected abstract group is therefore the symmetric group on four letters. This identification is now formalized in the production module

```
CGJteamLab/Coxeter/CoxeterRelations3DS4.lean
```

The proof is deliberately separated into two logically independent halves: faithfulness of the geometric action and surjectivity onto the full permutation group.

6.26 The Abstract Four-Vertex Type

The four vertices are represented by the finite inductive type

```
CoxeterA3Vertex
```

with constructors corresponding to A, B, C, D. For a fixed tetrahedral frame T, the map

```
CoxeterA3Vertex.point
```

sends the abstract vertices to the ambient spatial points

$$v_A \mapsto A, \qquad v_B \mapsto B, \qquad v_C \mapsto C, \qquad v_D \mapsto D.$$

Noncoplanarity of the frame immediately implies pairwise distinctness of the four ambient vertices. This is packaged as

```
CoxeterA3TetrahedralFrame.vertices_pairwise_ne
CoxeterA3Vertex.point_injective
```

and is the first bridge between the abstract permutation action and the spatial geometry.

6.27 The Three Adjacent Transpositions

Three explicit permutations of the abstract vertex type are defined:

```
sigma1
sigma2
sigma3
```

Their actions are

$$\sigma_1 = (v_A\, v_B), \qquad \sigma_2 = (v_B\, v_C), \qquad \sigma_3 = (v_C\, v_D).$$

The geometric reflection formulas already proved in the tetrahedral frame give the intertwining theorems

```
r1_point
r2_point
r3_point
```

which state, for every abstract vertex v,

$$r_i(\mathrm{point}(v)) = \mathrm{point}(\sigma_i(v)).$$

Thus the three geometric generators induce exactly the three standard adjacent transpositions on the four vertices.

6.28 The Generated Geometric Subgroup

The set of geometric generators is

```
geometricGenerators
```

and the subgroup of all ambient point permutations generated by them is

```
generatedGroup
```

that is,

$$G_T = \langle r_1, r_2, r_3 \rangle \leq \mathrm{Perm}(\texttt{Geo.Point}).$$

The predicate

```
RealizesVertexPerm
```

records that an ambient permutation g and an abstract vertex permutation σ are intertwined by the map `point`.

A closure induction proves

```
generated_realizes_vertex_perm
```

namely that every $g \in G_T$ realizes some permutation of the four vertices. Uniqueness follows from injectivity of `point`:

```
realized_vertex_perm_unique
```

This makes it possible to select the unique induced permutation

```
vertexPerm
```

and prove the multiplicative laws

```
vertexPerm_one
vertexPerm_mul
```

The result is the canonical homomorphism

```
vertexActionHom
```

with mathematical form

$$\rho_T : G_T \longrightarrow \mathrm{Perm}(\texttt{CoxeterA3Vertex}).$$

6.29 Faithfulness from Four-Point Rigidity

The injectivity of ρ_T is the point at which the geometric work of the previous sections returns.

First one proves, again by closure induction, that every element of `generatedGroup` preserves ambient segment congruence:

```
generated_preserves_congruence
```

Hence every $g \in G_T$ can be repackaged as a synthetic spatial isometry:

```
generatedIsometry
```

Suppose now that g lies in the kernel of ρ_T. The theorems

```
fixes_A_of_vertexAction_eq_one
fixes_B_of_vertexAction_eq_one
fixes_C_of_vertexAction_eq_one
fixes_D_of_vertexAction_eq_one
```

show that g fixes the four tetrahedral vertices.

The previously established rigidity theorem

```
HilbertSpaceIsometry3D.eq_refl_of_fix_noncoplanar4
```

then applies directly: a congruence-preserving equivalence fixing four noncoplanar points is the identity on the whole ambient space. Therefore

$$\ker \rho_T = \{1\}.$$

This is packaged as

```
eq_one_of_vertexAction_eq_one
vertexActionHom_injective
```

The faithfulness proof is therefore genuinely synthetic. No cardinality argument, matrix representation, coordinate calculation, or classification of finite groups is used.

6.30 Surjectivity from Adjacent Transpositions

For surjectivity, the four abstract vertices are given an explicit `Fintype` structure. The set

```
abstractGenerators
```

contains $\sigma_1, \sigma_2, \sigma_3$. Each is identified literally with an `Equiv.swap`:

```
sigma1_eq_swap
sigma2_eq_swap
sigma3_eq_swap
```

and consequently each satisfies `Equiv.Perm.IsSwap`.

The subgroup generated by these three swaps acts transitively on the four vertices. This is proved constructively in

```
abstractGenerators_closure_isPretransitive
```

using the chain

$$v_A \leftrightarrow v_B \leftrightarrow v_C \leftrightarrow v_D.$$

At this point the development uses the general Mathlib theorem

```
closure_of_isSwap_of_isPretransitive
```

from `Mathlib.GroupTheory.Perm.ClosureSwap`: a transitive permutation group generated by transpositions is the full symmetric group. Hence

$$\langle \sigma_1, \sigma_2, \sigma_3 \rangle = \mathrm{Perm}(\texttt{CoxeterA3Vertex}).$$

The three geometric generators are then packaged as elements

```
generatedR1
generatedR2
generatedR3
```

of G_T, and their images are proved to be

```
vertexAction_generatedR1
vertexAction_generatedR2
vertexAction_generatedR3
```

that is,

$$\rho_T(r_1) = \sigma_1, \qquad \rho_T(r_2) = \sigma_2, \qquad \rho_T(r_3) = \sigma_3.$$

Therefore the range of ρ_T contains the closure of the three adjacent transpositions, hence the full symmetric group. The final surjectivity statement is

```
vertexActionHom_surjective
```

6.31 The Final Isomorphism $G_T \cong S_4$

Injectivity and surjectivity are combined by

```
MulEquiv.ofBijective
```

to define

```
generatedGroupMulEquivS4
```

with type

$$G_T \simeq_{\text{grp}} \text{Perm}(\texttt{CoxeterA3Vertex}).$$

Since `CoxeterA3Vertex` has exactly four elements, the target is the full symmetric group on four letters, conventionally denoted S_4.

The generator images are recorded explicitly:

```
generatedGroupMulEquivS4_r1
generatedGroupMulEquivS4_r2
generatedGroupMulEquivS4_r3
```

and the final proposition-level endpoint is

```
generatedGroup_isomorphic_S4
```

Thus for every tetrahedral Coxeter frame T,

$$\boxed{\langle r_1, r_2, r_3 \rangle \cong S_4.}$$

Combined with

```
coxeter_A3_tetrahedral_frame_exists
```

this closes the first higher-rank realization unconditionally.

6.32 Final Axiom Audit

The final group-theoretic endpoint is audited by

```
#print axioms Geometry.CoxeterA3Vertex.generatedGroup_isomorphic_S4
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Quot.sound]
```

This is exactly the same axiom profile as the geometric A_3 existence theorem. In particular, the passage from the tetrahedral frame to S_4 introduces no new geometric or group-theoretic axiom.

6.33 Production Merge and Linter Cleanup for the S_4 Layer

The S_4 identification was first developed incrementally through five test modules. The complete proof was then merged into

```
CGJteamLab/Coxeter/CoxeterRelations3DS4.lean
```

without importing any of the test files. The production module imports only the completed $3D$ existence layer and the general Mathlib result on subgroups generated by transpositions.

After the merge the remaining linter warnings were removed by narrowing section-variable scopes, replacing an unnecessary `simpa`, and using an ordinary local `let` for the pretransitivity witness. The final production build is therefore clean.

6.34 A Wyler Flat Reinterpretation of the Same A_3 Frame

The preceding construction proves the A_3 relations globally and then identifies the generated group with S_4. The production module

```
CGJteamLab/Coxeter/CoxeterA3Wyler.lean
```

adds a different kind of information. It does not replace the tetrahedral construction, the global rigidity argument, or the S_4 identification. Instead, it asks how the already constructed three-dimensional mirror configuration decomposes when expressed in the language of generated flats, meets, joins, and explicit two-dimensional sections.

This distinction is important. The original A_3 proof shows that the three spatial reflections satisfy the required relations on the entire ambient point space. The Wyler layer explains where the pairwise Coxeter relations live geometrically.

For a pair of reflecting planes there are two natural pieces of geometry:

$$\boxed{\text{common fixed flat} \quad + \quad \text{active planar section.}}$$

In E^3, the common fixed flat of two distinct mirror planes is a line. The nontrivial action of the pair can then be read in a suitable plane transverse to that line. This is the three-dimensional prototype of the higher-dimensional pattern used later in E^4, where a pair of hyperplanes has a fixed 2-flat and an active 2-flat.

The Wyler layer therefore serves two purposes at once. It gives a second, structural reading of the already completed A_3 configuration, and it provides a low-dimensional laboratory for the flat and normal-section calculations needed in the nonvisual E^4 case.

6.35 The Mirror Meets as Spatial Flats

Write the three mirror planes of a tetrahedral frame T as

$$\pi_1, \qquad \pi_2, \qquad \pi_3.$$

Recall their defining geometry:

- π_1 is the perpendicular-bisector plane of AB, contains C, D, and has midpoint foot F_1;
- π_2 is the perpendicular-bisector plane of BC, contains A, D, and has midpoint foot F_2;
- π_3 is the perpendicular-bisector plane of CD, contains A, B, and has midpoint foot F_3.

The first flat calculation concerns the distant pair. Both F_1 and F_3 lie in $\pi_1 \cap \pi_3$, and they are distinct. The production theorem

```
wyler_pi1_meet_pi3_eq_span_F1_F3
```

states

$$\boxed{\mathrm{carrier}(\pi_1) \cap \mathrm{carrier}(\pi_3) = \mathrm{Span}\{F_1, F_3\}.}$$

Thus the common line of the two distant mirrors is not merely produced existentially: it is identified with the flat generated by the two opposite edge midpoints.

The adjacent pairs are packaged by

```
wyler_pi1_meet_pi2_line_through_D
wyler_pi2_meet_pi3_line_through_A
wyler_adjacent_mirror_meets
wyler_all_three_mirror_meets
```

and the three pairwise intersections are finally collected as actual line carriers by

```
wyler_each_mirror_pair_meets_in_line
```

Thus one obtains three lines

$$q_{12} = \pi_1 \cap \pi_2, \qquad q_{23} = \pi_2 \cap \pi_3, \qquad q_{13} = \pi_1 \cap \pi_3.$$

The theorem

```
wyler_three_meet_lines_pairwise_ne
```

proves that these three meet lines are pairwise distinct. This is the first point at which the mirror arrangement becomes a genuine incidence configuration rather than merely three unrelated pairwise intersections.

6.36 Join Reconstruction of the Three Mirrors

Once the three meet lines are available, the mirror planes can be recovered from them by joins. The theorem

```
wyler_A3_mirror_join_meet_calculus
```

packages the complete calculation:

$$\boxed{\begin{aligned} \pi_1 &= q_{12} \vee q_{13}, \\ \pi_2 &= q_{12} \vee q_{23}, \\ \pi_3 &= q_{23} \vee q_{13}. \end{aligned}}$$

Here the equalities are equalities of carriers produced through the three-dimensional Hilbert/Wyler join interface. No coordinates, normal vectors, or dimension formula are used.

This gives a compact incidence skeleton for the entire A_3 mirror arrangement:

$$\begin{array}{ccc} & \pi_2 & \\ q_{12} & & q_{23} \\ \pi_1 & q_{13} & \pi_3. \end{array}$$

The important point is not the diagram itself but its reconstruction property. The three lines determine the three mirrors pairwise by join. Consequently the flat structure already remembers the combinatorial shape of the A_3 mirror system before any Coxeter word is evaluated.

6.37 Meet Lines as Exact Common Fixed Loci

The next step connects incidence with the reflection transformations.

A point fixed by reflection in a plane is exactly a point of that plane. Therefore a point fixed simultaneously by two mirror reflections is exactly a point of the intersection of the two mirror carriers. The production theorems

```
wyler_pi1_pi2_meet_eq_commonFixed12
wyler_pi2_pi3_meet_eq_commonFixed23
wyler_pi1_pi3_meet_eq_commonFixed13
```

make this statement extensional for all three pairs.

The combined theorem

```
wyler_A3_meet_lines_eq_commonFixed_loci
```

therefore identifies

$$\boxed{q_{ij} = \mathrm{Fix}(r_i) \cap \mathrm{Fix}(r_j)}$$

at the level of point sets.

This is the precise fixed-flat interpretation of the Wyler meet lines. They are not merely intersections of planes chosen for incidence reasons: they are exactly the loci on which both generators act trivially.

The theorem

```
wyler_A3_flat_Coxeter_package
```

collects the incidence and fixed-point information into one package. This is the static part of the pairwise geometry. The Coxeter exponent itself is read from a complementary active section.

6.38 The First Adjacent Pair: Active Slice ABC

For the pair (π_1, π_2), choose the plane

$$\sigma_{12} = \text{plane}(A, B, C).$$

Noncoplanarity of the tetrahedral frame guarantees that this is a genuine plane and that neither mirror coincides with it.

Inside σ_{12}, the two mirror traces are

$$\pi_1 \cap \sigma_{12} = CF_1, \qquad \pi_2 \cap \sigma_{12} = AF_2.$$

The theorem

```
wyler_adjacent12_active_slice_lines
```

constructs these two carrier lines directly as plane meets, and

```
wyler_adjacent12_active_slice_axes
```

packages them as reflection axes in

```
PlaneGeo Geo sigma
```

for the chosen slice.

The reduction from spatial plane reflection to planar line reflection is then verified on the relevant vertices:

```
wyler_adjacent12_r1_reduces_to_lineReflect_on_A
wyler_adjacent12_r2_reduces_to_lineReflect_on_B
wyler_adjacent12_both_generators_planar_on_vertices
```

Geometrically, the two trace lines are two medians of the triangle ABC:

$$CF_1 \quad \text{and} \quad AF_2.$$

The metric regularity of this triangle is not added as a new tetrahedral axiom. It is recovered from the already established spatial reflection isometries.

Since r_2 fixes A and exchanges B, C,

$$AC \cong AB.$$

Since r_1 fixes C and exchanges A, B,

$$CA \cong CB.$$

Hence ABC is equilateral inside the slice. This is a useful structural fact: the equilateral triangle needed by the rank-two $p = 3$ theorem is forced by the mirror symmetries already present in the A_3 frame.

To match the existing planar theorem, the triangle is read in the order

$$(A, C, B).$$

Then

$$AF_2$$

is the median from the first vertex and

$$CF_1$$

is the median from the second. The theorem

```
wyler_adjacent12_planar_period_three_axes_match_traces
```

applies the established equilateral-median reflection theorem and matches the returned planar axis carriers with the two Wyler traces by line uniqueness.

The final natural-order statement is

```
wyler_adjacent12_traces_exact_period_three
```

and gives

$$\boxed{\mathrm{ExactPeriod}(\pi_1 \cap \sigma_{12}, \pi_2 \cap \sigma_{12}) = 3.}$$

Thus the exponent $m_{12} = 3$ is visible entirely inside an explicit two-dimensional section.

6.39 The Second Adjacent Pair: Active Slice BDC

The pair (π_2, π_3) is handled by the same mechanism, now in the plane

$$\sigma_{23} = \mathrm{plane}(B, D, C).$$

The mirror traces are

$$\pi_2 \cap \sigma_{23} = DF_2, \qquad \pi_3 \cap \sigma_{23} = BF_3.$$

Again the required equilateral geometry is derived from spatial reflection isometries rather than assumed independently.

Reflection in π_3 fixes B and exchanges C, D, hence

$$BC \cong BD.$$

Reflection in π_2 fixes D and exchanges B, C, hence

$$DB \cong DC.$$

Therefore the ordered triangle

$$(B, D, C)$$

is equilateral. Its two relevant medians are precisely

$$BF_3 \quad \text{and} \quad DF_2.$$

The complete construction is packaged by

```
wyler_adjacent23_traces_exact_period_three
```

and yields the natural-order statement

$$\boxed{\text{ExactPeriod}(\pi_2 \cap \sigma_{23}, \pi_3 \cap \sigma_{23}) = 3.}$$

Consequently the two adjacent Coxeter labels arise by the same local mechanism:

$$m_{12} = 3, \qquad m_{23} = 3.$$

The proof is not a second global rigidity argument. It is a reduction of each adjacent mirror pair to the already established planar $p = 3$ reflection theorem.

6.40 The Distant Pair: A Normal Slice and Period Two

The distant pair (π_1, π_3) has different geometry.

Its common fixed line is already known exactly:

$$q_{13} = \pi_1 \cap \pi_3 = \text{Span}\{F_1, F_3\}.$$

A plane containing q_{13} would not be useful as an active section: both mirror traces would then contain the same fixed direction. Instead, the construction uses a plane transverse to the common fixed line.

Let l_{AB} denote the carrier of the edge AB. Because F_1 is the midpoint foot for the perpendicular-bisector plane π_1,

$$l_{AB} \perp \pi_1 \quad \text{at } F_1.$$

At the same time $A, B \in \pi_3$, hence

$$l_{AB} \subset \pi_3.$$

The opposite edge carrier l_{CD} lies in π_1 and is the normal carrier associated with π_3. Through F_1, inside π_1, the construction draws a line s parallel to l_{CD}. This is obtained from the established Book XI parallel-through-a-point infrastructure used in Proposition XI.12.

Now let N be the plane generated by the intersecting lines

$$s \quad \text{and} \quad l_{AB}.$$

The two exact mirror traces are then proved to be

$$\boxed{\pi_1 \cap N = s, \qquad \pi_3 \cap N = l_{AB}.}$$

Moreover, since $l_{AB} \perp \pi_1$ at F_1 and $s \subset \pi_1$ passes through F_1,

$$l_{AB} \perp s.$$

The ambient perpendicularity is transferred to PlaneGeo(N) by the standard line-perpendicularity bridge. The two traces are packaged as a value of

```
ReflectionAxesPerpendicular
```

and the existing planar $p = 2$ theorem gives exact period two.

The complete result is

```
wyler_distant13_normal_slice_exact_period_two
```

which records simultaneously

$$\pi_1 \cap \pi_3 = \mathrm{Span}\{F_1, F_3\},$$

the two exact traces in N, their perpendicularity, and

$$\boxed{\mathrm{ExactPeriod}(\pi_1 \cap N, \pi_3 \cap N) = 2.}$$

Thus the distant exponent is not obtained from an equilateral triangle. It is the perpendicular-axis case in a genuine normal section.

6.41 The Local Coxeter Diagram from Flat Calculus

The capstone theorem of the production module is

```
wyler_A3_local_Coxeter_diagram
```

It packages the three local slice calculations into one statement:

$$\boxed{m_{12} = 3, \qquad m_{23} = 3, \qquad m_{13} = 2.}$$

More explicitly, it returns:

- a plane σ_{12} whose exact π_1, π_2 traces have reflection-product period 3;
- a plane σ_{23} whose exact π_2, π_3 traces have reflection-product period 3;
- a normal plane N whose exact π_1, π_3 traces are perpendicular and have reflection-product period 2;
- the distant fixed-flat identity $\pi_1 \cap \pi_3 = \mathrm{Span}\{F_1, F_3\}$.

The resulting architecture can be summarized as

$$\boxed{\begin{array}{c} \text{three spatial mirror planes} \\ \Downarrow \\ q_{12}, q_{23}, q_{13} \text{ by meet} \\ \Downarrow \\ \pi_1 = q_{12} \vee q_{13},\ \pi_2 = q_{12} \vee q_{23},\ \pi_3 = q_{23} \vee q_{13} \\ \Downarrow \\ q_{ij} = \text{exact common fixed locus of } r_i, r_j \\ \Downarrow \\ \text{active planar slices} \\ \Downarrow \\ m_{12} = 3, \quad m_{23} = 3, \quad m_{13} = 2. \end{array}}$$

This is the flat-calculus form of the A_3 Coxeter diagram.

6.42 Relation to the Original Global A_3 Proof

The Wyler layer should not be confused with a replacement of the earlier proof.

The original production theorem for the global Coxeter relations proceeds by the action on the four tetrahedral vertices and then uses synthetic four-point rigidity to promote equality on those anchors to equality of ambient spatial isometries. That argument proves relations of the actual three-dimensional plane reflections on every point of the ambient space.

The Wyler argument has a different purpose. It isolates, for each pair of generators, the geometry responsible for the corresponding Coxeter label. The comparison is therefore

global A3 proof	Wyler reinterpretation
ambient plane reflections	mirror meets and joins
vertex permutation	exact common fixed flat
four-point rigidity	active PlaneGeo slice
$(r_i r_j)^{m_{ij}} = 1$	planar exact period m_{ij}

The two arguments are complementary.

The global proof establishes the higher-rank realization and supports the subsequent identification with S_4. The flat proof explains why each edge of the Coxeter diagram carries its particular label and exposes a decomposition pattern that survives in higher dimension.

In this sense the A_3 case now has two distinct levels of understanding:

$$\boxed{\text{global realization} \quad + \quad \text{local fixed-flat/active-slice decomposition.}}$$

6.43 Production Status of the Wyler Layer

The Wyler development was first explored incrementally and then consolidated into the single production module

```
CGJteamLab/Coxeter/CoxeterA3Wyler.lean
```

without importing any of the experimental test files.

The final production module contains the complete chain

$$\text{meet/span} \Longrightarrow \text{join reconstruction} \Longrightarrow \text{common fixed loci} \Longrightarrow \text{active slices} \Longrightarrow (3,3,2).$$

The module builds cleanly on its own and is imported by the project root. After integration, a full

```
lake build
```

completed successfully with 2108 jobs.

No coordinates, vectors, inner products, Gram matrices, determinants, numerical angle measure, cosine, or trigonometry are introduced by this layer. Its additional language is incidence-theoretic—generated flats, meets, joins, carriers—together with the already established synthetic metric notions of midpoint, perpendicularity, congruence, and reflection.

The exact transitive axiom profile of the new capstone theorem may be recorded separately by a future `#print axioms` audit. The present production merge introduces no new axiom declaration specific to the Wyler A_3 module.

6.44 What is Complete and What Comes Next

The complete $3D/A_3$ stage now contains

- reflection in a plane;
- spatial reflection isometry;
- a tetrahedral A_3 frame;
- global Coxeter relations;
- braid relations and distant-generator commutation;
- the Wyler meet/join skeleton of the three mirror planes;
- exact identification of the three meet lines with common fixed loci;
- active planar slices realizing the local exact periods $m_{12} = 3$, $m_{23} = 3$, $m_{13} = 2$;
- synthetic existence of a regular tetrahedron;
- synthetic existence of the complete A_3 reflection frame;
- the canonical action of the generated group on four vertices;
- faithfulness from synthetic four-point rigidity;
- surjectivity from adjacent transpositions;
- the final group identification $\langle r_1, r_2, r_3 \rangle \cong S_4$.

There is therefore no remaining A_3 existence, relation, group-identification, or flat-decomposition checkpoint in the present program.

The algebraic part of the proof already points toward the family

$$A_{n-1} \longrightarrow S_n.$$

For n abstract vertices, the simple Coxeter generators should act as

$$(1\,2), \quad (2\,3), \quad \ldots, \quad (n-1\,n),$$

and the general Mathlib transposition-closure theorem is designed precisely for this situation. What remains genuinely geometric is to construct the corresponding higher-dimensional reflection frames synthetically and to prove the appropriate rigidity theorem needed for faithfulness.

Beyond type A, the broader Coxeter problem remains unchanged: one seeks a synthetic criterion ensuring that a family of reflecting hyperplanes realizes a prescribed Coxeter matrix simultaneously and controls the finiteness of the resulting group.

6.45 Conclusion

The first higher-rank Coxeter case is now complete both geometrically and group-theoretically. The geometric architecture is

$$\boxed{\begin{array}{c}\text{Hilbert 3-space}\\ \Downarrow\\ \text{plane reflection}\\ \Downarrow\\ \text{spatial reflection isometry}\\ \Downarrow\\ \text{tetrahedral frame}\\ \Downarrow\\ \text{global Coxeter relations } A_3\\ \Downarrow\\ \text{synthetic tetrahedral existence,}\end{array}}$$

The complementary Wyler architecture is

$$\boxed{\begin{array}{c}\text{tetrahedral mirror frame}\\ \Downarrow\\ \text{meet lines } q_{12}, q_{23}, q_{13}\\ \Downarrow\\ \text{join reconstruction of } \pi_1, \pi_2, \pi_3\\ \Downarrow\\ \text{exact common fixed loci}\\ \Downarrow\\ \text{active two-dimensional sections}\\ \Downarrow\\ m_{12} = 3, \quad m_{23} = 3, \quad m_{13} = 2.\end{array}}$$

This second architecture does not alter the global proof; it reveals its pairwise geometric content and supplies the fixed-flat/active-section pattern needed for higher-dimensional Coxeter work.

while the group-identification architecture is

$$\boxed{\begin{array}{c}G_T = \langle r_1, r_2, r_3\rangle\\ \Downarrow\\ \rho_T : G_T \to \mathrm{Perm}\{A, B, C, D\}\\ \Downarrow\\ \text{faithful by four-point rigidity}\\ \Downarrow\\ \text{contains } (AB), (BC), (CD)\\ \Downarrow\\ \text{surjective onto the full symmetric group}\\ \Downarrow\\ G_T \cong S_4.\end{array}}$$

No coordinate model is used in the geometric proof. The essential mechanisms are the local planar slices `PlaneGeo`, the spatial Hilbert congruence interface, Euclid Book XI for the interaction between lines and planes, and four-point spatial rigidity. The final algebraic closure step uses the abstract theory of permutation groups generated by transpositions.

This completes the synthetic A_3 chapter at three complementary levels: global spatial relations, group identification with S_4, and local Wyler fixed-flat/active-slice decomposition. It establishes the first fully realized higher-rank Coxeter group in CGJteamLab and the first completed low-dimensional model of the flat-calculus architecture later used in nonvisual dimension.

Chapter 7

Dimension-Free Incidence: the Smith–Wyler Calculus and the Hilbert E^3 Bridge

7.1 Purpose

The dimension-free part of the project was introduced in order to separate two logically different issues.

First, Hilbert three-space has a genuinely three-dimensional incidence axiom: two distinct planes with a common point have a second common point. In the formalization this is the field

```
HilbertSpaceIncidence.plane_second_common_point
```

of the spatial Hilbert incidence package.

Second, the Smith–Wyler flat calculus needs an incidence principle that continues to make sense in arbitrary dimension. Ambient plane–plane intersection cannot play that role: in E^4, two two-dimensional planes may meet in only one point. The dimension-free replacement is Smith I.5, equivalently the local Wyler I.7 / LP4 intersection principle used in the generated-flat calculus.

The present chapter records the production result that connects these two levels:

$$\text{Hilbert } E^3 \Longrightarrow \text{Smith–Wyler dimension-free incidence.}$$

The bridge is a theorem, not a new axiom.

7.2 The dimension-free incidence interface

The generic interface is

```
HilbertDimensionFreeIncidence
```

and packages the following incidence data:

```
two_points_on_each_line
three_noncollinear_on_plane
plane_through
plane_unique
line_in_plane
smith_i5
```

The first five clauses are the ordinary line–plane incidence skeleton. The last clause is the dimension-free Smith replacement for the three-dimensional Hilbert plane-intersection axiom.

This interface is intentionally retained. It is not merely a historical artifact of the E^3 development. It is the correct abstract hypothesis for the Smith–Wyler theory in arbitrary dimension, and in particular it is the layer that can be reused in the corrected E^4 development.

What changes in dimension three is its logical status: it can now be derived from the ordinary Hilbert spatial incidence axioms.

7.3 Three points are always coplanar in Hilbert E^3

The production bridge begins with

```
hilbert3D_three_points_coplanar
```

which proves that every triple of ambient points belongs to some spatial plane.

If the three points are noncollinear, the spatial plane-through-three-points axiom applies directly. If they are collinear, one chooses a point off their common line and constructs a plane through that line and the external point. The original three points then lie in the resulting plane.

This step is small but conceptually important. It allows the Smith I.5 configuration to be placed inside the ordinary Hilbert E^3 incidence geometry without adding any new plane-existence principle.

7.4 Smith I.5 from Hilbert I.7

The central theorem of the bridge is

```
hilbert3D_smith_i5
```

Its proof converts the dimension-free Smith configuration into a three-dimensional plane-intersection argument.

Let p be the distinguished point in the Smith configuration. Construct two planes

$$\alpha = \mathrm{plane}(p, p_0, q_0), \qquad \beta = \mathrm{plane}(p, p_1, q_1).$$

There are two cases.

7.4.1 Distinct planes

If $\alpha \neq \beta$, Hilbert I.7 supplies a second common point $q \neq p$. Since q lies on both planes, it satisfies the two coplanarity requirements of Smith I.5 immediately.

7.4.2 Coincident planes

If $\alpha = \beta$, the common plane by itself does not provide a second intersection plane. Choose a point R outside α, and construct a second plane γ through p and R. The planes α and γ are distinct and share p, so Hilbert I.7 again produces a second common point $q \neq p$. Since $q \in \alpha$, the required Smith coplanarity conclusions follow.

Thus the dimension-free Smith I.5 principle is a theorem of the ordinary three-dimensional Hilbert incidence axioms.

7.5 The production bridge

The entire package is assembled by

```
hilbert3D_to_dimension_free_incidence
```

with signature of the form

```
theorem hilbert3D_to_dimension_free_incidence
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo] :
    HilbertDimensionFreeIncidence Geo
```

No spatial order, congruence, Euclidean parallelism, metric structure, orthogonality, reflection theory, or continuity principle occurs in the hypotheses.

This is the exact formal meaning of the statement that the Smith–Wyler incidence layer is not an additional geometric assumption in E^3.

7.6 From Smith I.5 to Steinitz exchange

The generic dimension-free development remains independent of the three-dimensional bridge. Its production chain is

$$
\begin{aligned}
\text{Smith I.5} &\Longrightarrow \text{Wyler I.7 / LP4} \\
&\Longrightarrow \text{local Veblen–Young P3} \\
&\Longrightarrow \text{Wyler plane-cone flatness} \\
&\Longrightarrow \text{Wyler one-point generation} \\
&\Longrightarrow \text{generated-flat exchange} \\
&\Longrightarrow \text{SmithSpanExchange.}
\end{aligned}
$$

The relevant production modules include

```
HilbertDimensionFreeSmithCore.lean
HilbertDimensionFreeWyler.lean
HilbertDimensionFreeProjective.lean
HilbertDimensionFreeCone.lean
HilbertDimensionFreeSmithExchange.lean
```

The end-to-end E^3 consequence is recorded separately in

```
CGJteamLab/Hilbert3DSmithExchange.lean
```

by

```
theorem hilbert3D_implies_smithSpanExchange
    [H : HilbertIncidence Geo]
    [HP : HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo] :
    SmithSpanExchange Geo
```

The proof first constructs an ordinary local proof term

```
let D : HilbertDimensionFreeIncidence Geo := ...
```

and then passes D explicitly to the generic Smith–Wyler exchange theorem. No global type-class instance is installed.

7.7 Why the generic layer is not removed

The implication

$$\text{Hilbert } E^3 \Longrightarrow \text{HilbertDimensionFreeIncidence}$$

does not make the dimension-free layer redundant.

The direction of abstraction is

$$\boxed{\text{Hilbert } E^3 \longrightarrow \text{dimension-free Smith incidence} \longrightarrow \text{Wyler–Smith flat calculus.}}$$

The middle layer must remain visible because it is the reusable dimension-independent theory. In E^3 it is justified by the spatial Hilbert axioms; in E^4 and higher dimensions it must be supplied by a different ambient incidence analysis.

In particular, the dimension-free modules must not import the three-dimensional bridge. That would reverse the dependency direction and would incorrectly bake Hilbert I.7 into a theory whose purpose is precisely to avoid dimension-specific plane-intersection assumptions.

7.8 Audit of the three-dimensional consumers

The subsequent assumption audit searched the production E^3, Book XI, Wyler, and Coxeter sources for class-valued Hilbert, Smith, and Wyler hypotheses.

The result is structurally clean. The three-dimensional Coxeter and Book XI layers use the ordinary Hilbert hierarchy:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

The apparently suspicious occurrences of planar classes such as `HilbertCongruence`, `HilbertOrder`, or `HilbertEuclideanPlane` inside files with three-dimensional names are local planar helper theorems. They are applied inside induced plane geometries and are not additional ambient E^3 assumptions.

No separate three-dimensional construction class for normals, perpendicular planes, reflections, regular tetrahedra, or the A_3 frame remains in the production assumption boundary. Those constructions are represented by definitions, structures, or derived theorems.

7.9 Axiomatic status

The logical status of the completed incidence bridge can therefore be summarized as

$$\boxed{\text{Hilbert } E^3 \text{ incidence} \Longrightarrow \text{Smith I.5} \Longrightarrow \text{HilbertDimensionFreeIncidence} \Longrightarrow \text{SmithSpanExchange.}}$$

The bridge introduces no new geometric axiom and no new global instance. The only dimension-specific ingredient is the original Hilbert spatial incidence theory itself.

This conclusion is independent of the separate arbitrary-p existence problem for rank-two Coxeter periods. The explicit principle

```
HilbertFiniteEqualAngleHalfTurnDivisionExistence
```

remains the current assumption boundary for uniform finite half-turn division. It belongs to the metric/congruence existence problem for arbitrary p, not to the incidence-theoretic Smith–Wyler bridge proved in this chapter.

7.10 Dependency diagram

The final dependency picture for the present checkpoint is

Hilbert spatial incidence in E^3

$\Downarrow$

`hilbert3D_smith_i5`

$\Downarrow$

`hilbert3D_to_dimension_free_incidence`

$\Downarrow$

Smith–Wyler dimension-free flat calculus

$\Downarrow$

`dimensionFreeIncidence_implies_smithSpanExchange`

$\Downarrow$

`hilbert3D_implies_smithSpanExchange`.

This closes the incidence-theoretic passage from the classical Hilbert three-dimensional axioms to the abstract exchange language used by the Smith–Wyler part of the project.

Chapter 8

The Four-Dimensional Coxeter System A_4: Synthetic Hyperplane Reflections and S_5

8.1 Overview

This chapter records the first completed nonvisual Coxeter test in the development: the type-A_4 reflection system in a corrected synthetic four-dimensional Euclidean setting.

The principal production modules are organized through

```
CGJteamLab/Coxeter/E4Geometry.lean
CGJteamLab/Coxeter/CoxeterA4Smith.lean
CGJteamLab/Coxeter/CoxeterA4S5.lean
```

The first module collects the reusable corrected synthetic E^4 geometry. The second proves the geometric A_4 reflection relations for a fixed `CoxeterA4SmithSimplexFrame`. The third performs the final group identification and proves that the subgroup generated by the four geometric reflections is isomorphic to S_5.

The four geometric generators realize the standard adjacent transpositions on five distinguished vertices,

$$(0\,1), \qquad (1\,2), \qquad (2\,3), \qquad (3\,4),$$

and therefore generate the full symmetric group

$$\boxed{W(A_4) \cong S_5.}$$

with the geometric infrastructure decomposed into smaller modules rather than concentrated in one four-dimensional monolith.

The development is important for two distinct reasons.

First, it provides the next Coxeter system after the completed A_3 construction. Four generators must satisfy

$$r_i^2 = 1 \qquad (1 \leq i \leq 4),$$

$$(r_1r_2)^3 = (r_2r_3)^3 = (r_3r_4)^3 = 1,$$

and

$$(r_1r_3)^2 = (r_1r_4)^2 = (r_2r_4)^2 = 1.$$

Equivalently, adjacent generators satisfy the length-three braid relation, while nonadjacent generators commute.

Second, E^4 is the first case for which geometric correctness can no longer be monitored by an ordinary spatial picture. The proof therefore tests whether the architecture extracted from the planar and three-dimensional cases is genuinely synthetic and dimension-aware.

The central lesson of the chapter is

$$\boxed{\text{global configuration: } E^4 \qquad \text{local Coxeter relation: } E^2.}$$

The ambient dimension determines the configuration of hyperplanes and their intersections, but the metric interaction of a pair of reflecting hyperplanes is reduced to a canonical two-dimensional normal section.

8.2 Why A_4 is the First Decisive Test

The A_3 realization still admits direct geometric intuition. One can picture a tetrahedron, a reflecting plane, a common line of two planes, and a normal section.

None of these pictures controls E^4 globally.

A hyperplane in E^4 is three-dimensional, while the intersection of two independent hyperplanes is typically a two-dimensional flat. Thus the familiar three-dimensional description

$$\text{common axis} + \text{normal plane}$$

is no longer the correct ambient picture.

For two reflecting hyperplanes $H_i, H_j \subset E^4$, the structurally relevant decomposition is instead

$$\boxed{A_{ij} = H_i \cap H_j}$$

together with a normal two-plane

$$\boxed{P_{ij}.}$$

Both are two-dimensional. The first is fixed pointwise by the two reflections; the second contains the active reflection geometry.

This is the first place where the implementation is forced to distinguish what is genuinely ambient from what is merely local.

8.3 The Architectural Correction Before A_4

An early four-dimensional route attempted to reuse the existing three-dimensional incidence hierarchy as an ambient interface.

That approach was rejected.

The problem was structural rather than tactical: an ambient four-dimensional geometry must not inherit a class whose axioms encode three-dimensional rigidity or three-dimensional incidence behavior.

The corrected route therefore begins from dimension-free incidence and builds the E^4 specialization only after the dimension-independent objects have been separated from the $3D$-specific ones.

At production level the beginning of the route is represented by modules such as

```
CGJteamLab/HilbertDimensionFreeIncidence.lean
CGJteamLab/HilbertDimensionFreeFlats.lean
CGJteamLab/HilbertDimensionFreeExchange.lean
CGJteamLab/HilbertDimensionFreeSmithCore.lean
CGJteamLab/HilbertDimensionFreeWyler.lean
CGJteamLab/HilbertDimensionFreeProjective.lean
CGJteamLab/HilbertDimensionFreeCone.lean
CGJteamLab/HilbertDimensionFreeSmithExchange.lean
```

and the corrected E^4 layer is then exposed through

```
CGJteamLab/Coxeter/E4Geometry.lean
```

as a stable production entry point.

The purpose of this file is architectural: users of the four-dimensional Coxeter layer should not need to know the historical sequence of test and repair files from which the production API was extracted.

8.4 The Corrected E^4 Incidence and Metric Stack

The stable four-dimensional development is deliberately layered.

Its first stages separate incidence, order, congruence, and Euclidean structure:

```
E4Incidence
E4Order
E4Congruence
E4Euclidean
```

This separation is mathematically important.

Incidence describes how points, lines, planes, and higher flats are situated. Order controls betweenness and sidedness. Congruence supplies the metric relation on segments and angles. Euclidean structure supplies the parallel and perpendicular consequences needed for reflection geometry.

None of these layers should smuggle a three-dimensional ambient theorem into E^4.

The resulting dependency direction is

$$\boxed{\text{dimension-free incidence} \Longrightarrow E^4 \text{ incidence} \Longrightarrow E^4 \text{ order} \Longrightarrow E^4 \text{ congruence} \Longrightarrow E^4 \text{ Euclidean geometry.}}$$

This is the point at which the A_4 development becomes a test of the architecture rather than merely a test of individual lemmas.

8.5 Normal Geometry of a Hyperplane

Reflection in a hyperplane requires a synthetic substitute for the usual vector-normal construction.

The production chain therefore isolates a normal core and a marked two-dimensional metric model:

```
E4NormalCore
E4MarkedPlaneMetric
E4NormalUniqueness
```

The relevant geometric pattern is the same one that appears in lower dimensions, but it is formulated without an ambient scalar product.

For a hyperplane H and a point P, the construction identifies a perpendicular carrier through the appropriate foot $F \in H$. The reflected point P' is then characterized by the midpoint condition

$$F \text{ is the midpoint of } PP'.$$

The essential requirement is uniqueness. Without uniqueness, reflection remains only a geometric relation; with uniqueness it becomes a function.

Thus the same reconstruction pattern survives from line reflection and plane reflection:

$$\boxed{\text{reflection relation} \Longrightarrow \exists! \Longrightarrow \text{function} \Longrightarrow \text{involution} \Longrightarrow \text{equivalence.}}$$

8.6 Hyperplane Reflection as a Geometric Transformation

The corresponding production modules include

```
E4HyperplaneReflectionCore
E4HyperplaneReflectionEquiv
```

and later

```
E4HyperplaneReflectionIsometry
```

The first layer constructs the reflected point and proves the basic symmetry properties. The second packages the transformation as an equivalence of the ambient point type. The final isometry layer proves that segment congruence is preserved.

The order of these stages is intentional.

No transformation is postulated as primitive. The point map is extracted from a synthetic midpoint–perpendicular characterization, and only then is the transformation shown to have the expected metric behavior.

This is the same design principle used successfully for line reflection and plane reflection in earlier chapters.

8.7 Intersection of Two Reflecting Hyperplanes

For two distinct reflecting hyperplanes H and K, the development next isolates their common geometry.

The stable modules include

```
E4HyperplaneIntersection
E4HyperplaneCommonPlane
E4NormalSection
E4NormalParallel
```

The common intersection supplies the fixed block of the pair.

In E^4, the expected dimension count is

$$\dim(H \cap K) = 2.$$

The active geometry is supplied by a second plane generated by suitable normal directions.

Thus a pair of hyperplane reflections is decomposed into

$$\boxed{\text{fixed 2-flat} + \text{active 2-flat.}}$$

This decomposition replaces the three-dimensional intuition of a common axis and a normal plane.

8.8 The Normal Section

The normal-section construction is the geometric center of the A_4 argument.

The relevant production layers include

```
E4HyperplanePerpendicularFrame
E4NormalSectionNormal
E4ReflectionPlaneInvariance
E4NormalGeometry
```

```
E4NormalSectionPlanarSetup
E4NormalSectionReflectionRestriction
```

The purpose of this chain is to establish three facts.

First, the active normal section is a genuine two-dimensional geometric carrier.

Second, each ambient hyperplane reflection preserves that section.

Third, after restriction to the section, the ambient hyperplane reflection is the already understood planar line reflection.

Schematically,

$$\boxed{r_H|_P = s_h, \qquad r_K|_P = s_k,}$$

where $h = P \cap H$ and $k = P \cap K$ are the two reflection axes inside the normal section P.

This is the decisive reduction:

$$\boxed{\text{pair of hyperplane reflections in } E^4 \Longrightarrow \text{pair of line reflections in } E^2.}$$

No numerical angle is introduced.

8.9 Why the Pairwise Coxeter Relations Remain Planar

Once the two ambient reflections have been restricted to their active normal section, the metric content of the Coxeter relation is no longer four-dimensional.

For adjacent mirrors the planar section has the equilateral-median configuration already studied in Chapter 4. Hence the corresponding product has period three:

$$(r_i r_{i+1})^3 = 1.$$

For nonadjacent mirrors the relevant planar axes are perpendicular, so the product has period two:

$$(r_i r_j)^2 = 1, \qquad |i - j| > 1.$$

The conceptual gain is substantial. The E^4 development does not reprove a new metric theorem for every pair of hyperplanes. It proves a restriction theorem once, then reuses the established planar Coxeter geometry.

Thus the pairwise Coxeter matrix

$$m_{ij} = \begin{cases} 1, & i = j, \\ 3, & |i - j| = 1, \\ 2, & |i - j| > 1 \end{cases}$$

is realized through local two-dimensional sections.

8.10 Transport Back to the Ambient Geometry

A local planar relation is not yet a global equality of ambient transformations.

The next production layer therefore develops the transport properties needed to move reflection information through lines, planes, and hyperplanes:

```
E4HyperplaneReflectionCarrierCandidate
E4HyperplaneReflectionIncidence
E4HyperplaneReflectionLineTransport
E4HyperplaneReflectionPlaneTransport
E4HyperplaneReflectionCarrierIndependence
E4HyperplaneReflectionHyperplaneTransport
E4HyperplaneReflectionOrderTransport
```

These modules are not cosmetic infrastructure.

They express the fact that the reflected image of a geometric carrier is again the correct geometric carrier, and that this statement does not depend on an arbitrary auxiliary representation.

This is what allows a relation first verified inside one normal section to interact coherently with the full ambient configuration.

8.11 Ambient Triangle Transport

Two further modules provide synthetic metric transport in the ambient four-dimensional setting:

```
E4AmbientTriangleSAS
E4AmbientTriangleSSS
```

Their role is analogous to the spatial SAS machinery used in the three-dimensional A_3 chapter.

The point is not to construct a separate analytic metric in E^4. Instead, congruence information is transported through triangles using synthetic SAS and SSS principles.

These results are especially useful when the same metric configuration is seen in different local carriers.

They complete the bridge between local planar reflection geometry and ambient hyperplane geometry.

8.12 The A_4 Reflection Pattern

The geometric relation module is

```
CGJteamLab/Coxeter/CoxeterA4Smith.lean
```

and the final group-identification module is

```
CGJteamLab/Coxeter/CoxeterA4S5.lean
```

The geometric A_4 pattern uses five distinguished vertices

$$A_0, A_1, A_2, A_3, A_4$$

and four reflecting hyperplanes

$$H_1, H_2, H_3, H_4.$$

The intended vertex action is the adjacent-transposition pattern

$$r_1 : \ A_0 \leftrightarrow A_1,$$

$$r_2 : \ A_1 \leftrightarrow A_2,$$

$$r_3 : \ A_2 \leftrightarrow A_3,$$

$$r_4 : \ A_3 \leftrightarrow A_4,$$

with the remaining distinguished vertices fixed by the corresponding reflection. This is the geometric realization of the standard type-A_4 generator pattern

$$(0\,1), \qquad (1\,2), \qquad (2\,3), \qquad (3\,4).$$

At relation level this gives

$$(r_1r_2)^3 = 1, \qquad (r_2r_3)^3 = 1, \qquad (r_3r_4)^3 = 1,$$

and

$$(r_1r_3)^2 = 1, \qquad (r_1r_4)^2 = 1, \qquad (r_2r_4)^2 = 1.$$

Equivalently,

$$r_1r_2r_1 = r_2r_1r_2,$$

$$r_2r_3r_2 = r_3r_2r_3,$$

$$r_3r_4r_3 = r_4r_3r_4,$$

while

$$r_1 r_3 = r_3 r_1, \qquad r_1 r_4 = r_4 r_1, \qquad r_2 r_4 = r_4 r_2.$$

The important point is that these are obtained from actual synthetic hyperplane reflections, not by postulating an abstract Coxeter presentation.

Since the four generators act on the distinguished vertices as the adjacent transpositions

$$(0\,1), \quad (1\,2), \quad (2\,3), \quad (3\,4),$$

their generated permutation group is

$$\langle (0\,1), (1\,2), (2\,3), (3\,4) \rangle = S_5.$$

Thus the synthetic reflection realization identifies the Coxeter group of type A_4 with the symmetric group on five letters:

$$\boxed{W(A_4) \cong S_5.}$$

8.13 The Vertex Action Homomorphism

The final identification is not obtained merely by observing that the four formal symbols look like adjacent transpositions. The Lean development constructs the geometric subgroup

```
CoxeterA4S5.generatedGroup
```

as the subgroup of ambient point permutations generated by the four actual hyperplane reflections. It then associates to every element of this subgroup the induced permutation of the five distinguished vertices:

```
CoxeterA4S5.vertexActionHom
```

The generators are sent to

$$\sigma_1 = (0\,1), \qquad \sigma_2 = (1\,2), \qquad \sigma_3 = (2\,3), \qquad \sigma_4 = (3\,4).$$

Since the adjacent transpositions generate the full symmetric group, the vertex-action homomorphism is surjective. In Lean this is recorded by

```
CoxeterA4S5.vertexActionHom_surjective
```

Thus surjectivity is purely group-theoretic once the geometric action of the four reflections on the five vertices has been established.

8.14 Five-Anchor Rigidity and Faithfulness

Surjectivity alone is not enough. To identify the geometric group with S_5, one must also prove that no nontrivial ambient reflection word acts trivially on all five distinguished vertices.

The key synthetic theorem is

```
CoxeterA4S5.simplexFrame_five_anchor_distance_unique
```

For a fixed `CoxeterA4SmithSimplexFrame` with vertices A, B, C, D, E, it says that if two ambient points X, Y have equal corresponding distances from all five anchors,

$$AX = AY, \quad BX = BY, \quad CX = CY, \quad DX = DY, \quad EX = EY,$$

then

$$X = Y.$$

This statement is not assumed as a primitive four-dimensional rigidity axiom. It is proved synthetically. If $X \neq Y$, let M be the midpoint of XY. From the simplex-frame incidence data one can choose three of A, B, C, D which, together with M, are not coplanar. The corrected four-dimensional perpendicular-bisector construction then produces a hyperplane through those three anchors and M which contains every point equidistant from X and Y. Since all five simplex vertices are assumed equidistant from X and Y, all five would lie in that one hyperplane, contradicting the nonhyperplanarity stored in the simplex frame. Hence $X = Y$.

This closes the boundary that had earlier been represented by the abstract class `Hilbert4DFiveAnchorDistanceU`
That class is not an assumption of the production theorem.

Every element of the generated reflection subgroup preserves segment congruence. Therefore, if its induced vertex permutation is the identity, the original point X and its image have equal distances from all five anchors. Five-anchor rigidity gives equality pointwise. This yields

```
CoxeterA4S5.vertexActionHom_injective
```

and hence faithfulness of the action on the five distinguished vertices.

8.15 The Final Multiplicative Equivalence

Combining injectivity and surjectivity gives the production equivalence

```
CoxeterA4S5.generatedGroupMulEquivS5
```

and the public theorem

```
CoxeterA4S5.generatedGroup_isomorphic_S5
```

whose mathematical content is

$$\boxed{\langle r_1, r_2, r_3, r_4 \rangle \cong S_5}$$

for every fixed `CoxeterA4SmithSimplexFrame` satisfying the stated synthetic E^4 interfaces. The auxiliary theorem

```
CoxeterA4S5.S5_card
```

records the familiar numerical consequence

$$|S_5| = 120.$$

8.16 What Changed from the Original A_4 Program

The earlier design notes correctly identified A_4 as the first nonvisual benchmark, but several implementation choices were sharpened by the completed proof.

Most importantly, the successful route did not proceed by creating a single large "Hilbert $4D$" hierarchy modeled directly on the old three-dimensional hierarchy.

Instead, the development separated

$$\text{dimension-free incidence}$$

from

$$E^4\text{-specific ambient consequences}$$

and from

$$\text{local } 2D \text{ metric reasoning.}$$

This distinction is now one of the main mathematical results of the formalization effort.

The implementation also confirms that the local Coxeter relation should not be globalized by brute-force five-point rigidity unless such rigidity is independently useful.

For pairwise relations the more informative mechanism is the normal section.

8.17 Production Dependency Architecture

At a coarse level the stable dependency chain is

$$\boxed{\begin{array}{c}
\text{dimension-free incidence} \\
\Downarrow \\
E4Incidence \\
\Downarrow \\
E4Order \\
\Downarrow \\
E4Congruence \\
\Downarrow \\
E4Euclidean \\
\Downarrow \\
\text{normal geometry} \\
\Downarrow \\
\text{hyperplane reflection} \\
\Downarrow \\
\text{normal section} \\
\Downarrow \\
\text{planar restriction} \\
\Downarrow \\
\text{reflection isometry and transport} \\
\Downarrow \\
\text{ambient SAS/SSS} \\
\Downarrow \\
\texttt{CoxeterA4Smith} \\
\Downarrow \\
\text{vertex action on five anchors} \\
\Downarrow \\
\text{five-anchor rigidity} \\
\Downarrow \\
\texttt{CoxeterA4S5}.
\end{array}}$$

The root project does not import the entire chain individually.

Instead, the stable geometry is collected through

```
CGJteamLab/Coxeter/E4Geometry.lean
```

while the Coxeter application is split into the geometric relation module and the final group-identification module:

```
CGJteamLab/Coxeter/CoxeterA4Smith.lean
CGJteamLab/Coxeter/CoxeterA4S5.lean
```

This separation mirrors the mathematics:

$$\boxed{E^4 \text{ geometry} \neq A_4 \text{ application.}}$$

8.18 Workshop Files and Production Files

The A_4 development required a long sequence of experimental files, including many `test`, `fix`, and `refactor` stages.

These files record the proof search and remain useful as workshop history, but they are not part of the production dependency graph.

The production rule is therefore:

$$\boxed{\text{production modules do not import workshop modules.}}$$

The final import audit was performed specifically to remove remaining dependencies on historical files.

This distinction is particularly important in a development of this size. Without it, an apparently stable theorem can remain coupled to accidental intermediate definitions.

8.19 Comparison with the A_3 Mechanism

The A_3 and A_4 developments share the same broad synthetic philosophy but differ in their globalization mechanism.

For A_3:

$$\boxed{\text{local spatial computation} + \text{four-point rigidity} \Longrightarrow \text{global relation.}}$$

For A_4, the decisive pairwise mechanism is instead:

$$\boxed{\text{fixed intersection flat} + \text{active normal section} + \text{planar Coxeter theorem.}}$$

This is a better prototype for higher dimension.

The pairwise relation depends only on a two-dimensional active block, while the ambient dimension is carried by the surrounding flat structure.

Thus A_4 provides the first evidence that the Coxeter relation itself does not become metrically more complicated as the ambient dimension increases.

8.20 Axiom and Architecture Audit

The principal audit question for this chapter is not merely whether the final module compiles.

The stronger questions are:

1. Does the corrected E^4 route avoid importing a three-dimensional ambient incidence class?
2. Are planar Hilbert theorems used only inside explicit two-dimensional carriers?
3. Is hyperplane reflection reconstructed synthetically rather than postulated as a primitive map?

4. Are reflection restrictions to normal sections explicit?
5. Are production modules independent of historical `test/fix/refactor` files?
6. Is the A_4 application separated from the reusable E^4 geometry layer?

The completed production refactor was organized precisely to make these questions answerable at the module level.

The public theorem names are now frozen and the final `#print axioms` audit has been performed. Each of

```
CoxeterA4S5.generatedGroup_isomorphic_S5
CoxeterA4S5.generatedGroupMulEquivS5
CoxeterA4S5.vertexActionHom_injective
CoxeterA4S5.simplexFrame_five_anchor_distance_unique
```

depends transitively only on

```
propext
Classical.choice
Geometry.bookZero_nullSegment1
Geometry.bookZero_nullSegment2
Quot.sound
```

No `Hilbert4DFiveAnchorDistanceUniqueness` axiom appears in the final signature or axiom audit. In particular, the faithfulness step is now a proved synthetic consequence of the corrected E^4 incidence and perpendicular-bisector machinery rather than an abstract rigidity boundary.

8.21 Conclusion

The A_4 development is the first completed step in the project whose ambient geometry cannot be directly visualized.

Its significance is therefore larger than the addition of one more Coxeter diagram.

The proof confirms a reusable structural principle:

$$\boxed{\text{higher-dimensional reflection geometry} = \text{ambient flat structure} + \text{local planar metric geometry.}}$$

For the Coxeter relations themselves, the active metric problem remains two-dimensional.

The role of E^4 is to supply the correct hyperplanes, their fixed intersections, the normal sections, and the transport needed to lift the local result back to the ambient configuration.

The four hyperplane reflections therefore realize not only the Coxeter relations of type A_4, but the standard adjacent-transposition action of the full symmetric group S_5.

Consequently,

$$\boxed{W(A_4) \cong S_5.}$$

This makes A_4 the first serious validation of the intended route toward higher-rank synthetic Coxeter systems.